WATER +
PREVENTIVE
MAINTENANCE

Water & Preventive Maintenance Foundation

水及預防性維護基礎課程

AST Guidebook 講師指南

COFFEE
TECHNICIANS
PROGRAM

Specialty
Coffee
Association

Water & Preventive Maintenance Foundation

AST Guidebook V1.0 (English)

水及預防性維護基礎課程 講師指南 V1.0 (中文版)

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Note: This guidebook replaces previous curriculum documents. This guidebook is only available to authorized trainers licensed in Water and Preventive Maintenance Foundation. Please do not share this document with other ASTs or learners

注意：先前課程文件已由本參考手冊取代。相關內容僅供水與預防性維護基礎課程培訓師使用．請勿與其他講師或學員分享。

1. General Information 一般資訊

Course Information

Course Length: Minimum 9 hours

Minimum course hours are met through time spent in the following ways:

- The AST delivering live instruction in-person or through distance learning.
- The AST conducting informal and formal assessments. (Note: This does not include time spent by the learner taking the online written exam.)
- The learner engaging in activities that introduce and/or reinforce the theoretical or practical course objectives.

Prerequisites: None

課程資訊

課程時間長度: 最短九小時

最短授課時數可透過以下方式達成：

- 由 AST 提供實體授課或遠距教學
- 由 AST 進行非正式或正式評測（請注意：學員參加線上筆試的時間不列入時數）
- 學員參加理論類或實務類的入門和/或進階活動

先修課程：無

Written Exam Information:

Total Number of Questions on Online Written Exam: 30 (worth one point each)

Total Time Allowed for Online Written Exam: 32 minutes

Passing Score (Online Written Exam): 70%

Practical Exam Information:

Total Time Allowed for Practical Exam: 60 minutes

Passing Score (Practical Exam): 70% = 70 points correct / 100 total exam points

Learner must pass each section of the Practical Exam

筆試資訊

線上筆試總題數：30 題（每題 1 分）

線上筆試時間限制： 32 分鐘

及格分數（線上筆試）： 70%

實作測驗資訊

實作測驗時間總長：60 分鐘

及格分數（實作測驗）：70% = 得分 70 / 總分 100

學生必需通過實務操作考試的每個部分

2. Course Description

Description

Water and Preventive Maintenance Foundation is an introductory course which covers key concepts of water treatment systems and preventive maintenance of coffee machines and certain practical skills required to safely apply this knowledge.

At the end of the course, the learner will be able to:

- Identify typical components and describe the function of common water treatment systems
- Define key water parameters and describe their potential effects on equipment function
- List water quality recommendations
- Describe expected effects of common water treatment methods on water quality
- Demonstrate measurement of incoming water to determine composition
- Demonstrate common steps for maintaining a water treatment system
- Identify typical wear components in a traditional espresso machine, batch brewer, and coffee grinder
- Describe the goals of a preventive maintenance program
- List the tools and supplies needed to perform preventive maintenance on coffee equipment
- Demonstrate simple preventive maintenance steps for a traditional espresso machine, batch brewer, and coffee grinder

2. 課程說明

說明

水與預防性維護的基礎課程,是一堂入門課程,含概了水處理系統、咖啡機器的預防性維護及為了安全的實施這些知識所需的相關實務技巧

在課程結束後，學生將能夠達成：

- 辨認一般水處理系統的基本組成零件並能夠描述其功能
- 定義主要水的參數讀值，並描述它們對設備功能的潛在影響
- 列出水質的建議
- 描述一般水處理方法在水質上的期望效果
- 示範如何對進水水源進行檢測以了解其成分組成
- 示範一般維護水處理設備的步驟
- 辨認標準義式咖啡機,批次沖煮及咖啡磨豆機所需標準耗材
- 描述預防性維護計畫的目標
- 列出實施咖啡設備的預防性維護所需要的工具及供應

3. Written Exam Question Distribution by Topic

The chart below sets forth key information regarding the online exam questions.

Question Pool: This is the number of questions **per topic** that are available to present to the learner during the online exam.

Questions Presented: This is the number of questions a learner will randomly receive **per topic** during the online exam. This number was determined by the creators for the purpose of ensuring that each section and topic of the course is weighted appropriately.

Section Weighting: Next to each section title is the percentage of the total exam represented by the questions in that section.

There are some topics/objectives that are not tested during the written or practical exam. The trainer should still address these topics/objectives as the information will be useful in future courses.

3. 按主題分類的筆試內容

有關線上測驗試題的重要資訊詳如下表

題庫：每個主題會用於學員線上測驗的試題數量。

考題：**每個主題**實際列入施測內容（隨機選出）的試題數量。該數量係由企劃者決定，旨在確保課程每個章節和主題的比重恰如其分。

章節比重：每個章節名稱右側為相關考題在整份測驗中的占比。部分主題/目標並未列入筆試或實作測驗。由於這些資訊將在未來課程中派上用場，因此培訓師仍應講授相關內容。

See grids on next two pages. 請參閱下兩頁的表格

Exam Sections & Topics 考試章節及主題	Question Pool 題庫	Questions Presented 考題	Exam Sections & Topics 考試章節及主題	Question Pool 題庫	Questions Presented 考題
1.01 | Section | Workplace Safety Practices | Pre-Course **1.01 |章節|工作場域安全實作|事前課程**			**1.04 | Section | Water Quality Recommendations | 7%** **1.04 |章節|水質建議|7%**		
1 Topic: Personal Responsibility 7.1.主題: 個人責任	Not Tested 未有考題		1 Topic: SCA Water Recommendations 1.主題:SCA 水質建議	3	1
2 Topic: Corporate & Manufacturing Oversight 2.主題: 公司及製造商的監管	Not Tested 未有考題		2 Topic: The SCA Water Chart 2.主題:SCA 水質圖	3	1
3 Topic: Clothing & Personal Protection Equipment 3.主題: 衣著及個人保護設備	Not Tested 未有考題		3 Topic: Manufacturer Water Recommendations 3.主題:製造商的水質建議	Not Tested 未有考題	
4. Topic: Reducing Environmental Risks 4.主題:減少環境風險	Not Tested 未有考題		**1.05 | Section | Water Treatment Methods | 13%** **1.05 | 章節 | 水處理方法 | 13%**		
5 Topic: Moving Equipment 5.主題:移動設備	Not Tested 未有考題		1 Topic: Water Treatment Methods 1.主題: 水處理方法	Not Tested 未有考題	
6 Topic: Applying the Correct Tools 6.主題:運用正確的工具	Not Tested 未有考題		2 Topic: Effects of Carbon Filtration 2.主題: 活性碳濾芯的功效	3	1
7 Topic: Mental Preparedness 7.主題: 心理準備	Not Tested 未有考題		3 Topic: Effects of Water Softening 3.主題: 水質軟化的功效	3	1
8 Topic: Safe Workshop Practices 8.主題: 安全工作場域的實務	Not Tested 未有考題		4 Topic: Effects of Reverse Osmosis Filtration 4.主題: 逆滲透過濾的功效	3	1
9 Topic: Recognizing Food Safety Standards 9.主題: 認知食品安全標準	Not Tested 未有考題		5 Topic: Effects of Remineralization 5.主題:水質回填的功效	3	1
1.02 | Section | Water Treatment System Components | 7%			**1.06 | Section | Measuring Water Composition | 10%**		

1.02 | 章節 |水處理設備零件 | 7%		
1 Topic: Sanitary Quick-change Cartridge System Components 1.主題:乾淨的快拆濾芯系統零件	3	1
2 Topic: Drop-in Cartridge System Components 2.主題: 置入式濾芯系統零件	Not Tested 未有考題	
3 Topic: Reverse-Osmosis System Components 3.主題: 逆滲透系統零件	3	1
4 Topic: Tank-style System Components 4.主題:桶式系統零件	Not Tested 未有考題	
1.03 | Section | Defining Key Water Parameters | 13% **1.03 | 章節 | 定義主要的水參數 | 13%**		
1 Topic: Water Composition 1.主題: 水的組成	Not Tested 未有考題	
2 Topic: Total Hardness and Alkalinity 2.主題:總硬度及鹼度	4	1
3 Topic: Temporary and Permanent Hardness 3.主題: 暫時及永久性硬度	3	1
4 Topic: Total Dissolved Solids Estimation 4.主題: 總溶解固體的估算	3	1
5 Topic: pH 5.主題:pH 值	Not Tested 未有考題	
6 Topic: Other Substances in Water 6.主題: 水中的其他物質	3	1

1.06 | 章節 | 測量水質組成 | 10%		
1 Topic: Water Testing Tools 1. 主題:水質測量工具	Not Tested 未有考題	
2 Topic: Using an Electrical Conductivity/TDS Meter 2.主題: 使用導電度計/TDS 儀	Tested on Practical 於實務測驗進行考試	
3 Topic: Using a Drop-Count Titration Kit 3. 主題: 使用滴定儀	3	1
4 Topic: Using a pH Meter 4. 主題:使用 pH 儀	3	1
5 Topic: Using Water Testing Strips 5.主題:使用水質檢測試紙	3	1
6 Topic: Using the SCA Water Chart 6.主題:使用 SCA 水質圖	Tested on Practical 於實務測驗進行考試	
1.07 | Section | Water Treatment System Maintenance | 7% **1.07 | 章節 | 水處理系統的維護 | 7%**		
1 Topic: Maintenance Requirements 1.主題: 維護需求	3	1
2 Topic: Maintaining a Cartridge System 2.主題: 濾芯系統的維護	3	1
3 Topic: Maintaining a Drop-in Cartridge System 3.主題: 置入式濾芯系統的維護	Tested on Practical 於實務測驗進行考試	

Exam Sections & Topics **測驗章節&主題**	**Question Pool** **題庫**	**Questions Presented**	**Exam Sections & Topics** **測驗章節&主題**	**Question Pool** **題庫**	**Questions Presented**

		考題
1.08 | Section | Preventive Maintenance Components | 17% **1.08 | 章節 | 預防性維護零件 | 17%**		
1 Topic: Espresso Machine Brewing Group Components 1. 主題:義式咖啡機沖煮頭零件	3	1
2 Topic: Espresso Machine Steam Valve Components 2. 主題:義式咖啡機蒸氣閥零件	3	1
3 Topic: Espresso Machine Steam Boiler Components 3. 主題: 義式咖啡機蒸氣鍋爐零件	3	1
4 Topic: Batch Brewer Components 4. 主題:滴濾式咖啡機零件	3	1
5 Topic: Coffee Grinder Components 5. 主題:咖啡磨豆機零件	3	1
1.09 | Section | Maintenance Strategies | 7% **1.09 | 章節 | 維護策略 | 7%**		
1 Topic: Preventive versus Reactive Maintenance 1.主題:預防性 V.S. 反應性維護	3	1
2 Topic: Recommended Maintenance Intervals 2. 主題:建議保養週期	3	1
1.10 | Section | Preventive Maintenance Tools and Supplies | 7% **1.10 | 章節 | 預防性維護的工具與零件 | 7%**		
1 Topic: Recommended Hand Tools 1.主題: 建議手動工具	3	1
2 Topic: Recommended Consumables 2.主題: 建議耗材	3	1

		考題
1.11 | Section | Performing Maintenance | 13% **1.11 | 章節 | 維護的實施 | 13%**		
1 Topic: Safety During a PM Service 1. 主題: 預防性維護服務的安全	3	1
2 Topic: Preventive Maintenance Checklist 2. 主題: 預防性維護檢核表	Tested on Practical 於實務測驗進行考試	
3 Topic: Espresso Machine PM Cleaning Steps 3.主題:義式咖啡機預防性保養的清潔步驟	3	1
4 Topic: Espresso Machine PM Parts Replacement 4.主題:義式咖啡機預防性保養的零件更換	Tested on Practical 於實務測驗進行考試	
5 Topic: Espresso Machine PM Inspection Steps 5.主題: 義式咖啡機預防性保養的檢查步驟	3	1
6 Topic: Batch Brew Preventive Maintenance 6.主題:滴濾式咖啡機的預防性維護	3	1
7 Topic: Espresso Grinder Preventive Maintenance 7.主題:義式磨豆機的預防性維護	Tested on Practical 於實務測驗進行考試	
Totals	91	30

4. Course Curriculum with Corresponding Online Written Exam Questions

The course curriculum is set forth below and is divided into Sections, Topics and Objectives. In some areas of the curriculum, the creators may have revised the curriculum in order to create a more logical, level-appropriate structure. Any revisions are noted in *2. Course Description and Updates.*

All online written exam questions were developed as an assessment for a specific objective. These questions have been grouped according to topic. All questions within a topic are considered the topic "pool." From this pool, a certain number of questions will be randomly selected and presented to the learner. If a particular topic has more than one objective, there is a possibility that the learner will not be tested on all objectives in the topic. This is due to the randomization of the questions from that topic.

4. 課程大綱與相關的線上筆試考題

課程大綱呈現如下,並區分成區段,主題及目標. 在大綱中的某些部分,為了讓整體結構更富有邏輯性及程度的適切性,創作者可能會修正大綱. 任何修正會在 2. 課程描述及更新中註明.

所有的線上筆試問題會因應特定學習目標的評估而設定.這些問題會因主題而分群. 在一個主題下的所有問題被認為是這個主題的試題庫. 從這個試題庫中,這些問題會在線上筆試中隨機的呈現給學生. 如果某一特定主題有超過一個的學習目標, 因為主題下的問題是隨機呈現的,學生可能不會在每一個學習目標都接受測驗.

1.01 | Section | Workplace Safety Practices
1.01|章節|工作場域安全實作 共

11 Topics11 個主題

Topic 主題	Objectives 目標	AST Notes 講師附註	Online Written Exam Questions	Resources
1.01.01 Personal Responsibility **個人責任**	A. State that workplace safety is a shared responsibility. B. State that the individual is ultimately responsible for their own personal safety. C. State that individuals share a responsibility for the safety of those around them. A 指出工作場域的安全是一個大家共同的責任 B.指出每個人需對自己的安全負起最終的責任 C.指出個人對他們週圍的安全是有共同責任的	Emphasize that workplace safety is shared responsibility and requires attention from all involved, but that each technician is responsible for keeping themselves and others around them from harm. Discuss the importance of compliance and intervention to the effectiveness of policies and procedures. 強調工作場域的安全是共同的責任,並且需要所有人的參與,但每個技師都有責任要保護他們自己及周圍的人使其遠離危險. 討論規範與干預對於有效執行政策與程序的重要性	**Not Tested.** 無測試	
1.01.02 Corporate and Manufacturing Oversight 公司及製造商的監管	A. List the relevant agencies that promote policies for the protection of workers. B. List the relevant agencies and certifications that exist to regulate safety features in equipment and tools. A. 列出推動政策來保護工作人員	Give examples of the relevant national and regulatory bodies. Examples include the U.S. Occupational Safety and Health Administration (OSHA), the Korean Occupational Safety and Health Agency (KOSHA), the European Agency for Safety and Health at Work (EU-OSHA), the Health and Safety Executive ministry in the UK (HSE) and the International Labor Organization (ILO). Provide information about state and local licensing for technicians performing certain duties and give references for more information.	**Not Tested.** 無測試	

	的相關機構 B. 列出監管設備及工具安全的相關機構及認證	Provide information about relevant equipment certifiers such Underwriters Laboratories (UL), Central European (CE), Engineering Test Lab (ETL), Canadian Standards Association (CSA), and others. 舉例說明相關國際及獨立機關. 這些例子包含了美國職業安全衛生署(OSHA), 韓國職業安全健康局(KOSHA), 歐盟職業安全衛生署(EU-OSHA), 英國衛生安全局(HSE), 國際勞工組織(ILO) 提供執行某些任務的技師所需的國家及當地的證照且給予進一步資訊的相關參考 提供相關設備證照的資訊,像保險實驗室(UL),歐盟認證(CE),工程測試實驗室(ETL),加拿大標準協會(CSA)以及其他		
1.01.03 Clothing and Personal Protection Equipment **衣著及個人保護設備**	A. List common personal protective equipment that reduces the risk of workplace injury B. List common considerations when choosing workplace clothing and accessories, including jewelry A. 列出一般減少工作場域傷害風險的個人保護性設備 B. 列出在選擇工作場域服裝及配件,包含首飾時的一般考量	Give examples of how certain tasks may require specific personal protection equipment for that application. Personal protection equipment that is common in most situations include: - eye protection - appropriately-sized work gloves - ear protection Technicians should pursue: - shoes that protect their toes - clothing that fits appropriately Technicians should avoid: - wearing loose clothing - wearing loose or obtrusive jewelry such as watches, bracelets or rings. - loose long hair 舉例說明某些任務如何需要特定的保護設備而加以應用 在多數狀況如下,個人保護設備是常見的 -眼睛保護	**Not Tested.** **無測驗**	

		-適當大小的工作手套 -耳朵保護 技師應該要貫徹: -可以保護腳趾的鞋子 -大小剛好恰當的衣著 技師應該要避免: -穿著過於寬鬆的衣著 -著戴寬鬆或突兀的首飾像是手錶、手鏈或戒指 -鬆鬆的長髮		
1.01.04 **Mental Readiness** **心理狀態**	A. Recognize mental distractions that can increase safety risk. B. Recognize when further training is necessary before attempting a test or repair. A. 認知心理上的分心會增加安全的風險 B. 認知在嘗試開始測試或修理前,進一步的訓練是必需的	Emphasize the important role that focus plays in maintaining a safe working environment. Discuss the different types of mental distractions that can decrease focus. These may include: - fatigue - daydreaming - worries or problems in personal life - conversations with co-workers - feeling hurried by undue pressure from customer/manager Hold a discussion about recognizing when a task goes beyond abilities or level of training. 強調在維護安全的工作環境中所扮演的重要角色 討論會降低專注的幾種心理上的分心. 他們包含了: -疲勞 -白日夢 -個人生活中的問題或煩惱 -同事間的聊天 -來自於客戶或管理者的不當壓力而使其感到匆忙倉促	**Not Tested.** **無測驗**	
1.01.05	A. Recognize the safety risks posed	List the types of chemical hazards that may be faced by a coffee	**Not Tested.**	

Chemical Hazards **化學危害**	by chemical hazards. A.辨認化學危害所產生的安全性風險	machine technician. Examples may include: - chemicals used for cleaning - chemicals used for descaling coffee equipment - chemicals used for lubrication Display the warning language and symbols that are used in the country or region of the student. 列出一個咖啡機技師可能會面臨不同型的化學危害 舉例包含 -清潔所使用的化學物質 -對咖啡設備除垢所使用的化學物質 -潤滑所使用的化學物質 展示學生的國家或地區所使用的警語及符號	**無測驗**	
1.01.06 **Fire Hazards** **火災危害**	A. Recognize the safety risks that can be posed by fire hazards in a workspace. B. State the source for information about flammable and explosive material. A. 辨識在工作場域中會產生火災危害的安全性風險 B. 指出易燃及爆炸性物質的相關資訊及來源	List the types of fire hazards that may be faced by a coffee machine technician. Examples may include: - structural fires - electrical fires - chemical fires Display the warning language and symbols that are used in the country or region of the student. Discuss the importance of emergency response plans for fire suppression or evacuation. Share resources for further study. 列出一個咖啡機技師可能會面臨的火災危害類型 舉例包含 -結構性火災 -電氣火災 -化學火災 展示學生的國家或地區所使用的警語及符號 討論火災發生時,滅火及疏散警急應變計畫的重要性.分享資源以達到更進一步的研究	**Not Tested.** **無測驗**	
1.01.07	A. State best practices for safely	Share the considerations and best practices for moving coffee	**Not Tested.**	https://www.osha.gov/

Moving Equipment **設備的移動**	moving large and/or heavy equipment. A 指出安全移動較大或重的設備時所採用的最佳措施	equipment. Considerations should include: - The correct, neutral posture - The removal of tripping hazards for a clear path - Developing and communicating a plan for the movement - Using the right number of people - Proper handholds Demonstrate tools used for the cartage of coffee machines 分享移動咖啡設備時的考量與最佳措施 考量應包含 -正確而中性的姿勢 -清除障礙物以保持乾淨清楚的動線通道 -發展並討論移動的計畫 -正確的人力使用 -搬運時有適當的提把處 示範搬運咖啡機所使用的工具		SLTC/etools/ electricalcontractors/ supplemental/ principles.html#lifting
1.01.08 **Applying Appropriate Tools** **使用恰當的工具**	A. State the importance of using the correct tool for the task at hand. A. 指出使用正確的工具來完成手邊工作的重要性	Discuss how using the wrong tool for a task can result in damage or injury. Discuss how the dulling of a tool that is designed to be sharp can result in damage or injury. Discuss how the condition of tools in the workplace can be regulated by state or national bodies. 討論在工作中使用錯誤的工具如何能導致災難與傷害 討論本應被設計為銳利的工具在鈍掉後如何能導致災難與傷害 討論國家或國際組織如何規範工作場域中工具的狀況	**Not Tested.**	
1.01.09 **Shopkeeping Practices**	A. List safety considerations for the workplace environment. A.列出工作場域環境中的安全性考量	Discuss considerations for organizing and operating a safe workplace environment. Considerations should include: - Proper lighting - Sure footing - Clear signage and markings for warnings and directions - Restriction of workstations (moveable chains, stanchions and	**Not Tested.**	

		similar.) - Restriction of unsupervised visitors - Clearly signage for emergency response - Lockout & tagout procedures - Clearly marked supplies - Clearly marker chemicals 討論組織及運作一個安全工作場域環境的考量 這些考量應該要包含: -正確的燈光 -腳下安全 -清楚易懂的警告標幟及方向指示 -工作站的管制(可移動的支柱、圍條及相關類似之物) -管制未受監督的訪客 -緊急應變的清楚指示 -安全上鎖掛牌程序 -清楚標識的零件 -給予化學物品清楚的標識		
1.01.10 Workstation Practices **工作站的實務操作**	A. List safety considerations for individual workstations. A 列出個人工作站的安全考量	Discuss considerations for organizing and operating a safe workstation. Considerations should include: - Proper lighting - A designated place for every tool and supply - Clearly marked supplies - Appropriate height working surface - Void of spills and debris - Equipped with cleaning materials 討論組織運作一個安全工作站的考量 這些考量應該要包含: -正確的燈光 -保留清楚標識的空間給每個工具及零件 -清楚標識的零件	**Not Tested.** **無測試**	

		-恰當高度的工作枱面 -保留會濺出或有碎片的空間		
1.01.11 **Respecting Food Safety Standards** **遵守食物安全標準**	A. List best practices that respect food safety standards while working in a food handling space. A. 列出在食物加工環境工作時,能達成遵守食物安全標準的最佳措施	Emphasize that technicians must be aware of the manufacturer's cleaning recommendations for each piece of equipment. Discuss the surfaces of a coffee machine that come in contact with the beverage product and the importance of cleanliness to the customer. Discuss how certain cleaning compounds, such as alkyl sulfonate, can have offensive tastes and are detectable at concentrations of 100 ppm. 強調技師必需清楚明白製造商對於設備中每一部分的清潔建議. 討論咖啡機中與飲品有所接觸的表面以及之於顧客端清潔的重要性 討論某些清潔化合物,像是烷基磺酸酯,會有討厭的味道,並且在濃度達到 100ppm 時是可以被察覺到的	**Not Tested.** **無測驗**	

1.02 | Section | Water Treatment System Components

1.02 | 章節 | 水處理設備零件 共

4 Topics4 個主題

Topic 主題	Objectives 目標	AST Notes 講師附註	Written Online Exam Questions 線上筆試問題	Resources 相關資源
1.02.01 **Quick-change Cartridge System Components** **快拆濾芯系統零件**	A. List common components of a quick-change cartridge water treatment system, including: - System head - Cartridge - Shut-off valve(s) - Pressure gage A.列出一般快拆過濾水處理系統的組成零件,包含: -卡式濾頭 -快拆卡式濾芯 -進水閥 -壓力計	`	**Question ID: 100010842451** A commonly used type of water filter system is shown in this picture. Which of the following terms best describes the system component designated by the arrow?  **System head** Quick-change cartridge Booster pump Shut-off valve 問題編號: **100010842451** 在照片中我們可以看到一組常見的水過濾系統. 下方哪一個用詞最適合用來描述箭頭所標出的零件 **卡式濾頭** 快拆卡式濾芯 加壓幫浦 進水閥	

			Question ID: 100010842452 A commonly used type of water filter system is shown in this picture. Which of the following terms best describes the system component designated by the arrow?  **Shut-off valve** System head Quick-change cartridge Booster pump 問題編號: **100010842452** 在照片中我們可以看到一組常見的水過濾系統. 下方哪一個用詞最適合用來描述箭頭所標出的零件 **進水閥** 卡式濾頭 快拆卡式濾芯 加壓幫浦	
			Question ID: 100010842453 A commonly used type of water filter system is shown in this picture. Which of the following terms best describes the system component designated by the arrow? 	

			Quick-change cartridge System head Booster pump Shut-off valve 問題編號: **100010842453** 在照片中我們可以看到一組常見的水過濾系統. 下方哪一個用詞最適合用來描述箭頭所標出的零件 **快拆卡式濾芯** 卡式濾頭 加壓幫浦 進水閥	
	B. Identify common components of a quick-change cartridge water treatment system B. 辦認一組快拆水處理系統的組成零件	Activity 1.02, See Appendix A - Activity Preparation Guide 活動 1.02 看附錄 A 活動準備指南	**This objective has an activity.** **這個教學目標有活動**	
1.02.02 **Drop-in Cartridge System Components** **置入式過濾系統零件**	A. List common components of a drop-in water treatment system, including: - System housing - O-ring - Shut-off valve(s) - Cartridge A. 列出一般置入式水處理系統的零件,包含: - 系統濾水器外殼 - O 形密封矽膠環 - 進水閥 - 濾芯	Discuss various configurations of components which may be used by different manufacturers 討論不同製造商所使用各式不同的零件配置與構造	**Not tested.** **無測驗**	

	B. Identify common components of a drop-in water treatment system B 辨認一組置入式過濾系統的組成零件	Activity 1.02, See Appendix A - Activity Preparation Guide 活動 1.02 看附錄 A 活動準備指南	**This objective has an activity.** **這個課程目標有活動**	
1.02.03 Reverse-Osmosis System Components **逆滲透過濾系統零件**	A. List common components of a reverse-osmosis water treatment system, including: - System head - Booster pump - Sediment filter - Carbon filter - Membrane - Blending valve - Storage tank A.列出逆滲透過濾處理系統的一般零件,包含: -系統上座 -加壓幫浦 -棉質濾芯 -活性碳濾芯 -過濾膜 -混水閥 -儲水桶	Discuss various configurations of components which may be used by different types of reverse osmosis systems and different manufacturers 討論不同製造商及不同的逆滲透過濾系統所使用各式不同的零件配置與構造	**Question ID: 100010842476** Which type of water filtration system often uses a booster pump and a membrane? **Reverse osmosis system** Water softener Carbon filter Remineralizing system 問題編號: **100010842476** 哪一種型態的水過濾系統通常會用到加壓幫浦和過濾膜? **逆滲透過濾系統** 軟水系統 活性碳濾芯 礦物質回填系統	
			Question ID: 100010842480 Which of the following components are **NOT** typically part of a reverse osmosis water treatment system? **Water softener cartridge** Carbon filter Booster pump Membrane 問題編號: **100010842480** 下方哪一個零件並未在經典的逆滲透過濾水處理系統中? **軟水濾芯**	

			活性碳濾芯 加壓幫浦 過濾膜	
			Question ID: 100010842485 Which of the following components is missing from this reverse osmosis system: 1. Booster pump 2. System head 3. Carbon Filter 4. ___________ 5. Storage Tank **Membrane** Softener cartridge Polyphosphate cartridge Decarbonizing cartridge 問題編號: **100010842485** 下方哪一個零件在逆滲透過濾系統中消失了: 1.加壓幫浦 2. 系統上座 3. 活性碳濾芯 4. 5. 儲水桶 **過濾膜** 軟水濾水器外殼 多磷酸塩濾芯 脫碳濾芯	
1.02.04 **Tank-style System Components** **桶式系統零件**	Identify a tank-style water treatment system 辨認桶式水處理系統		**Not tested.** **無測驗**	

1.03 | Section | Defining Key Water Parameters
1.03 | 章節 | 定義主要水的參數

6 Topics
六個主題

Topic 主題	Objectives 目標	AST Notes 講師重要	Written Online Exam Questions 線上筆試題目	Resources 相關資源
1.03.01 **Water Composition** **水的組成**	A. List typical sources of substances commonly dissolved in water, including: - the atmosphere - contact with rock and soil - pollution of surface water - municipal water treatment process - saltwater intrusion A.列出典型一般溶解在水中物質的來源,包含: -大氣 -與岩石及土壤的接觸 -地表水的污染 -自來水處理設備的過程 -沿海低窪地區的塩水入侵		**Not tested.** **無測驗**	The SCA Water Quality Handbook, pg. 10 SCA 水質手冊 第 10 頁
1.03.02 **Total Hardness and Alkalinity** **總硬度及鹼度**	A. Define total hardness as the combined quantity of calcium and Magnesium ions dissolved in water B. Define alkalinity as the acid buffering capacity of water and concentration of carbonate and bicarbonate compounds in the water	Alkalinity is sometimes called carbonate hardness AST should discuss that alkalinity differs from Alkaline - having a pH higher than 7. 鹼度有時被稱為碳酸鹽硬度	**Question ID: 100010842659** Which of the following parameters describes the combined quantity of calcium and magnesium ions dissolved in water? **Total hardness** Permanent hardness Alkalinity Total dissolved solids	The SCA Water Quality Handbook, pg. 12 SCA 水質手冊 第 12 頁

	C. State that the recommended unit of measure for total hardness and alkalinity is ppm $CaCO_3$ A.定義總硬度為溶解於水中鈣離子與鎂離子的總合 B 定義鹼度為水中緩衝酸的能力,為水中碳酸根及碳酸氫根的濃度	講師應討論鹼度與鹼性並不相同 鹼性為 PH 值大於 7	問題編號:100010842659 下面哪一個參數用以描述水中溶解鈣離子與鎂離子的總量? **總硬度** 永久硬度 鹼度 總溶解固體量	
			Question ID: 100010842661 Is the following statement true or false? Total hardness defines the quantity of dissolved calcium carbonate in water. **False** True 問題編號:100010842661 請問以下的論述正確與否? 總硬度定義為水中溶解碳酸鈣的量值? **錯誤** 正確	The SCA Water Quality Handbook, pg. 12 SCA 水質手冊 第 12 頁
			Question ID: 100010842662 Which of the following units is commonly used to quantify total hardness? ppm $CaCO_3$ German degrees Grains per Gallon **All of the above** 問題編號:100010842683 下面哪一個通常被用來作為總硬度測量的單位?	The SCA Water Quality Handbook, pg. 15 SCA 水質手冊 第 15 頁

			PPM CaCO3 德制硬度 克/加崙 **以上皆是**	
	D. List other common measurement units for total hardness and alkalinity, including: - German degrees - Grains per US Gallon D. 列出其他一般用來作為總硬度或鹼度測量的單位? -德制硬度 -克/美式加崙		**Question ID: 100010842664** Which of the following units is commonly used to quantify alkalinity? **ppm CaCO3** µS Degrees Celsius pH 問題編號:100010842664 下面哪一個通常被用來作為總鹼度測量的單位? **ppm 碳酸鈣** µS 攝氏溫度 pH	The SCA Water Quality Handbook, pg. 16 SCA 水質手冊 第 16 頁
1.03.03 **Temporary and Permanent Hardness** **暫時及永久性硬度**	A. Define Temporary and Permanent Hardness A. 定義暫時性及永久性硬度		**Question ID: 100010842665** Which of the following water parameters indicates the dissolved calcium and/or magnesium which **CAN** precipitate out of a sample of water when boiled? **Temporary hardness** Permanent hardness Total hardness Alkalinity 問題編號:100010842665 下面哪一個水的參數指的是在水被煮滾後,水中溶解會被沈澱析出的鈣及鎂? **暫時硬度**	The SCA Water Quality Handbook, pg. 47 SCA 水質手冊 第 47 頁

			永久硬度 總硬度 鹼度	
			Question ID: 100010842666 Which of the following water parameters indicates the dissolved calcium and/or magnesium which **CANNOT** precipitate out of a sample of water when boiled? **Permanent hardness** Temporary hardness Total hardness Alkalinity 問題編號: **100010842666** 下面哪一個水的參數指的是在水被煮滾後,水中溶解不會被沉澱析出的鈣及鎂? **永久硬度** 暫時硬度 總硬度 鹼度	
			Question ID: 100010842667 Which of the following best defines temporary hardness? **Dissolved calcium and/or magnesium which can precipitate out of a sample of water when boiled** Dissolved calcium and/or magnesium which cannot precipitate out of a sample of water when boiled Dissolved calcium and/or magnesium which occur as compounds other than carbonates in water The total quantity of dissolved calcium and/or magnesium	

			問題編號: **100010842667** 下方哪一個選項是暫時硬度的最佳定義? **水中溶解的鈣與鎂在水煮滾後會被沈澱析出** 水中溶解的鈣與鎂在水煮滾後不會被沈澱析出 水中溶解的鈣與鎂以其他非碳酸鹽化合物的形式存在 水中溶解的鈣與鎂的總量	
1.03.04 **Total Dissolved Solids Estimation** **總溶解固體的估算**	A. State that the electrical conductivity of water can be used to estimate the total dissolved solids (TDS) in water A. 指出水的導電度可用來估算水中的總溶解固體(TDS)含量	Handheld TDS meters actually measure the electrical conductivity of the sample which is then used to display TDS based on a conversion factor. 手持式的 TDS 測量儀實際上測量的是樣本的導電度,以其做轉換因子而成為 TDS 的概略值	**Question ID: 100010842668** What physical parameter of a water sample does a handheld TDS meter measure in order to produce a TDS reading? **Electrical conductivity** Total dissolved solids pH Refractive index 問題編號: **100010842668** 為了製造 TDS 的讀數,手持式 TDS 測量儀以水樣本中的哪一個物理參數來做測量? **導電度** 總溶解固體含量 pH 折射率	Water for Coffee, pg 56-58 咖啡中的水 56-58 頁
			Question ID: 100010842669 Is the following statement true or false? Handheld TDS meters measure the electrical conductivity of a liquid and display an estimate of the total dissolved solids in the sample. **True** False 問題編號: **100010842669** 以下的論述正確與否?	

			手持式 TDS 測量儀測量液體的導電度並顯示樣本總溶解固體含量的概略值 正確 錯誤	
			Question ID: 100010842670 Which of these easily measured parameters is often used to estimate the total dissolved solids in a water sample? **Electrical conductivity** pH Turbidity Hardness 問題編號: **100010842670** 下列這些容易測量的參數中,哪一個通常被用來估算樣本的水中總固體含量 **導電度** PH 值 濁度 硬度	
1.03.05 pH **pH 值**	A. Define pH A. 定義 PH 值		**Not tested.** **無測驗**	
1.03.06 **Other Substances in Water** **其他水中物質**	A. State that other substances are sometimes found in drinking water, including: - Sodium - Iron - Chlorine - Chlorides - odor-causing compounds - sediment		**Question ID: 100010842672** Which of the following substances is sometimes found dissolved in drinking water? Sodium Iron Odor-causing compounds **All of the above**	Equilibrium in Water and Coffee, pg 86 水與咖啡的平衡

	A. 指出飲用水中有時會找到的其他物質, 包含: -鈉 -鐵 -氯 -有味道的物質 -沈澱物		問題編號: **100010842672** 下方哪一個物質有時會被發現溶解於飲用水中 鈉 鐵 有味道的化合物 **以上皆是**	
			Question ID: 100010842673 Which of the following is an undesirable substance that may sometimes be found in municipal drinking water? **Sediment** Calcium Magnesium Bicarbonate 問題編號: **100010842673** 下方哪一個物質有時會被發現於供水系統的飲用水中,卻是一個令人討厭的物質 **沈澱物** 鈣 鎂 碳酸氫根	
			Question ID: 100010842675 Which of the following is an undesirable substance that may sometimes be found in municipal drinking water? **Odor-causing compounds** Alcohol Magnesium Bicarbonate	

			問題編號: **100010842675** 下列哪一個令人討厭的物質有時會出現在供水系統的飲用水中? 有味道的物質 酒精 鎂 碳酸氫根	

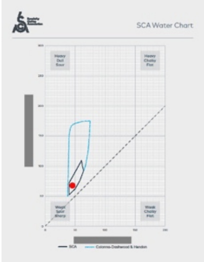
1.04 | Section | Water Quality Recommendations 1.04 | 章節 | 水質建議

3 Topics 三個主題

Topic 主題	Objectives 目標	AST Notes 講師附註	Written Online Exam Questions 線上筆試問題	Resources 相關資源
1.04.01 **SCA Water Recommendations** **SCA 水質建議**	A. Recognize that the SCA Water Quality Handbook recommends that water for coffee extraction should: - Be odor free and hygenic - Have a pH of between 6-8 - Be free of iron, lead, or other unsuitable or toxic compounds in significant amounts A. 辨識 SCA 水質手冊建議咖啡萃取用水應該: -無外來的味道且衛生 -PH 值在 6-8 之間 -沒有鐵、鉛或其他不恰當或明顯量	Legacy SCAA Water Quality Standard specified that Chlorine be limited to 0 mg/L and sodium limited to 10mg/L (for Superior Brew) 傳統 SCAA 水質標準明確指出氯含量的限制為零,而鈉含量為 10mg/L(為了更好的沖煮)	**Question ID: 100010842683** Which of the following characteristics does the SCA recommend that water for coffee brewing exhibit? Odor free Hygienic Free of iron, lead, or toxic compounds **All of the above** 問題編號:100010842683 下方的哪個特質符合 SCA 建議的咖啡沖煮用水? 無外來風味 衛生 無鐵、鉛或其他有毒物質 以上皆是	SCA Water Quality Handbook, pg. 74-76 SCA 水質手冊 74-76 頁

	質的有毒物質		**Question ID: 100010842685** What range of pH does the SCA recommend for water for coffee extraction? **6-8** 5-7 7-9 There is no recommendation 問題編號:100010842685 SCA 建議咖啡萃取用水的 PH 值範圍為何? **6-8** 5-7 7-9 沒有建議	
			Question ID: 100010842686 According to SCA Water Quality Handbook, which of the following substances is acceptable in moderate amounts in water for coffee brewing? **Calcium** Chlorine Iron Lead 問題編號:100010842685 根據 SCA 水質手冊,在咖啡沖煮用水中,下方哪一個物質適量存在是可以接受的? 鈣 氯 鐵 鉛	

1.04.02 **The SCA Water Chart** **SCA 水質圖**	A. Recognize that the SCA Water Chart represents recommended levels of total hardness and alkalinity for water for coffee brewing B. State that the SCA Water Chart recommendation minimizes the risk of limescale formation and corrosion in espresso machines and hot water boilers A. 辨識 SCA 水質圖代表了對咖啡沖煮使用水整體硬度及鹼度的建議數值 B. 指出 SCA 水質圖的建議會極小化義式咖啡機及熱水鍋爐中水垢的形成及管線的鏽蝕		**Question ID: 100010842689** Which **two (2)** of the following water parameters are shown on the SCA Water Chart? **Total hardness** **Alkalinity** Total dissolved solids (TDS) Permanent hardness 問題編號: **100010842689** 在下方水參數中,有哪二個出現在 SCA 的水質圖中? 總硬度 鹼度 總溶解固體含量(TDS) 永久硬度	
			Question ID: 100010842691 Which **two (2)** of the following potential equipment risks does the SCA Water Chart's Core Zone minimize? **Limescale formation** **Corrosion** Silicate deposits Iron deposits 問題編號: **100010842691** SCA 水質圖中的中心區域可以極小化下方潛在設備風險中的哪二個? 水垢的形成 鏽蝕 矽酸塩的沈澱 鐵的沈澱	

			Question ID: 100010842692 Which of the following is a risk of using the water with the parameters shown here on the SCA Water Chart?  **There is no risk** Limescale formation Pitting corrosion Silicate buildup 問題編號: **100010842692** 使用如圖所示 SCA 水質圖中的水質參數,會有什麼樣的風險? 無風險 結水垢 孔蝕 矽酸塩積聚沈澱	
1.04.03 Manufacturer Water Recommendations **製造商的水建議**	A. Recognize that equipment manufacturers provide recommendations for water composition, which should be followed. B. State that water containing levels of chlorides above manufacturer recommendations may result in corrosion in coffee equipment A. 應當辨識並遵守設備製商所提供		**Not tested.** **無測試**	

	的水質組成建議 B. 指出水中氯含量高於設備製造商的建議時可能會導致咖啡設備的鏽蝕			

1.05 | Section | Water Treatment Methods 1.05 | 章節 | 水處理的方法

5 Topics 五個主題

Topic 主題	Objectives 目標	AST Notes 講師附註	Written Online Exam Questions 線上筆試問題	Resources 相關資源
1.05.01 **Water Treatment Methods** **水處理的方法**	A. List commonly used water treatment methods, including: - carbon filtration - sediment filtration - water softening - reverse osmosis filtration - remineralization A. 列出通常會用來進行水處理的方法,包含: -活性碳濾芯 -棉質濾芯 -水質軟化 -逆滲透過濾 -礦物質回填		**Not tested.** **無測驗**	

1.05.02 **Effects of Carbon Filtration** **活性碳濾芯的功用**	A. List expected changes to water composition as a result of carbon filtration B. List aspects of water composition which are not expected to be changed as a result of carbon filtration A. 列出在經過活性碳濾芯後所期待的水質組成改變 B. 列出在經過活性碳濾芯後,無法改變的水質組成部分		**Question ID: 100010842696** Which of the following effects is expected if water is treated using a carbon filter? **Odor-causing compounds are removed** Significant reduction in total hardness Significant reduction in alkalinity Significant reduction in total dissolved solids 問題編號: **100010842696** 如果使用活性碳濾芯作為水處理的方法,下方哪個功效是被期待的? **去除有味道的物質** 在總硬度上的明顯下降 在總鹼度上的明顯下降 在總溶解固體含量上的明顯下降	
			Question ID: 100010842698 Which **two (2)** of the following effects are **not** expected if water is treated using a carbon filter? **Significant reduction in total hardness** **Significant reduction in alkalinity** Odor-causing compounds are removed Chlorine taste is removed 問題編號: **100010842698** 如果使用活性碳濾芯作為水處理的方法,下方哪二個功效是**不**被期待的? **在總硬度上的明顯下降** **在總鹼度上的明顯下降** 去除有味道的物質 去除氯的味道	

			Question ID: 100010842699 Which of the following effects is expected if water is treated using a carbon filter? Unpleasant taste is removed Significant reduction in total hardness Significant reduction in alkalinity Significant reduction in total dissolved solids 問題編號: **100010842699** 如果使用活性碳濾芯作為水處理的方法,下方哪個功效是被期待的? 去除不舒服的味道 在總硬度上的明顯下降 在總鹼度上的明顯下降 在總溶解固體含量上的明顯下降	
1.05.03 **Effects of Water Softening** **水質軟化的功效**	A. List expected changes to water composition as a result of water softening B. List aspects of water composition which are not expected to be changed as a result of water softening A. 列出在經過水質軟化濾芯後所期待的水質組成改變 B. 列出在經過水質軟化濾芯後,無法改變的水質組成部分	The SCA Water Chart should be used to visually represent the expected changes to total hardness and alkalinity during discussions 在討論時,SCA 水質圖應該用以視覺上觀察總硬度及鹼度所期待的改變	**Question ID: 100010842702** Which of the following effects is expected if water is treated using a water softener? Significant reduction in total hardness Odor-causing compounds are removed Significant reduction in alkalinity Significant reduction in total dissolved solids 問題編號: **100010842702** 如果使用水質軟化濾芯作為水處理的方法,下方哪個功效是被期待的? **在總硬度上的明顯下降** 去除有味道的物質 在總鹼度上的明顯下降 在總溶解固體含量上的明顯下降	

			Question ID: 100010842703 Which **two (2)** of the following effects are **NOT** expected if water is treated using a water softener? **Significant reduction in alkalinity** **Significant reduction in total dissolved solids** Significant reduction in total hardness Reduced risk of limescale buildup 問題編號: **100010842703** 如果使用水質軟化濾芯作為水處理的方法,下方哪二個功效是**不**被期待的? **在總鹼度上的明顯下降** **在總溶解固體物質上的明顯下降** 在總硬度上的明顯下降 減少水垢形成的風險	
			Question ID: 100010842704 Which of the following water treatment methods is typically used to reduce only the total hardness of water? **Water softening** Remineralization Carbon filtration Reverse osmosis 問題編號: **100010842704** 下方的哪一個水處理方法通常被只用來減少水中的總硬度 **水質軟化濾芯** 礦物質回填濾芯 活性碳濾芯 逆滲透過濾	

1.05.04 **Effects of Reverse Osmosis Filtration** **逆滲透過濾的功效**	A. List expected changes to water composition as a result of reverse osmosis filtration B. List aspects of water composition which are not expected to be changed as a result of reverse osmosis filtration C. State that controlled amounts of untreated water can be blended with reverse osmosis treated water prior to final filtration. Describe the expected changes to the water composition. A. 列出在經過逆滲透過濾系統後所期待的水質組成改變 B. 列出在經過逆滲透過濾系統後,無法改變的水質組成部分 C.指出將已比例控制的未處理水與逆滲透過濾系統後的處理水加以混合，再進行最終的過濾，描述在水質組成上所期待的改變。	The SCA Water Chart should be used to visually represent the expected changes to total hardness and alkalinity during discussions 在討論時,SCA 水質圖應該用以在視覺上代表在總硬度及鹼度所期待的改變	**Question ID: 100010842705** Which of the following effects is expected if water is treated using a reverse osmosis system? Significant reduction in total hardness Significant reduction in alkalinity Significant reduction in total dissolved solids **All of the above** 問題編號:**100010842705** 如果使用逆滲透過濾系統作為水處理的方法,下方哪個功效是被期待的? 在總硬度上的明顯下降 在鹼度上的明顯下降 在總溶解固體含量上的明顯下降 以上皆是	
			Question ID: 100010842706 Which of the following effects are **NOT** expected if water is treated using a reverse osmosis system? **Increase in total dissolved solids** Significant reduction in alkalinity Significant reduction in total dissolved solids Significant reduction in total hardness 問題編號: **100010842706** 如果使用逆滲透過濾系統作為水處理的方法,下方哪個功效是**不**被期待的? **增加總溶解固體含量** 在鹼度上的明顯下降 在總溶解固體含量上的明顯下降	

			在總硬度上的明顯下降	
			Question ID: 100010842707 Which of the following approaches may be used to increase the mineral level of water which has been treated using a reverse osmosis system? **Blending in some untreated water prior to final filtration** Ion exchange softening Decarbonization water softening Carbon filtration 問題編號: **100010842707** 下方的何種方法可用於在逆滲透過濾系統中增加水中的礦物質? **在最終過濾前與一些未處理的水加以混合** 離子交換以軟化水質 脫碳水質軟化 活性碳濾芯	
1.05.05 **Effects of Remineralization** **礦物質回填的功效**	A. List expected changes to water composition as a result of remineralization B. List aspects of water composition which are not expected to be changed as a result of remineralization A. 列出在經過礦物質回填系統後所期待的水質組成改變 B. 列出在經過礦物質回填系統後,無法改變的水質組成部分		**Question ID: 100010842708** Which of the following effects is expected if water is treated using remineralization? **Increased total hardness and alkalinity** Reduced total hardness and alkalinity Odor-causing compounds are removed Significant reduction in total dissolved solids 問題編號: **100010842708** 如果使用礦物質回填系統作為水處理的方法,下方哪個功效是被期待的? **增加總硬度及鹼度** 減少總硬度及鹼度 去除有味道的物質 在總溶解固體含量上的明顯下降	

			Question ID: 100010842709 Which **two (2)** of the following effects are **NOT** expected if water is treated using remineralization? **Reduction in total dissolved solids** **Odor-causing compounds are removed** Increased total hardness Increased alkalinity 問題編號: **100010842709** 如果使用礦物質回填系統作為水處理的方法,下方哪二個功效是**不**被期待的? **在總溶解固體含量上的明顯下降** **去除有味道的物質** 增加總硬度 增加鹼度	
			Question ID: 100010842710 Which of the following water treatment methods is typically used to increase the mineral content of water? **Remineralization** Water softening Carbon filtration Reverse osmosis 問題編號: **100010842710** 下方的哪一個水處理方法通常被用來增加水中的礦物質含量? **礦物質回填濾芯** 水質軟化濾芯 活性碳濾芯 逆滲透過濾	

1.06 | Section | Measuring Water Composition 1.06 | 章節 | 測量水質的組成

6 Topics 六個主題

Topic 主題	Objectives 目標	AST Notes 講師附註	Written Online Exam Questions 線上考試問題	Resources 相關資源
1.06.01 **Water Testing Tools** **水質測量工具**	A. List measurement tools commonly used for measuring water composition, including: - Electrical conductivity/TDS meters - Drop-count titration kits - pH meters - test strips B. Describe best practices for collecting and testing water samples A. 列出一般在測量水質組成所需工具,包含: -導電度計/TDS 測量儀 -滴定儀 -PH 儀 -檢測試紙 B. 描述收集及檢測水樣本的最佳措施		**Not tested.** **無測驗**	

1.06.02 **Using an Electrical Conductivity/TDS Meter** **使用導電度計/TDS 測量儀**	A. State limitations of electrical conductivity-based measurement of total dissolved solids A. 指出以導電度為基準來進行總溶解固體濃度測量的限制	The electrical conductivity of samples varies depending on the exact composition of the dissolved minerals as well as the temperature of the sample, so the true TDS of a sample can only be estimated using this method. Note that meters may use different conversion factors Estimated values can be useful if total hardness and alkalinity are known 樣本的導電度會因實際溶於水中的礦物質組成及溫度而有所不同,因此某一樣本真正的總溶解固體濃度僅能以此方法進行估算 需注意測量儀可能會用不同的換算因子 在總硬度及鹼度是可知的狀態下,總溶解固體濃度的概略值是有效的	**Not tested.** **無測驗**	Water for Coffee, pg 56-58
	B. Demonstrate proper usage of an Electrical Conductivity/TDS meter B. 示範正確使用導電度計/TDS 測量儀	Activity 1.06, See Appendix A - Activity Preparation Guide 活動 1.06 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	
1.06.03 **Using a Drop-Count Titration Kit** **使用滴定儀**	A. Recognize that drop count titration is a recommended method for assessing total hardness and alkalinity of a water sample A. 辨別滴定法是測量水質樣本的總		**Question ID: 100010842711** Which of the following testing methods is recommended for use by a coffee equipment technician to measure the total hardness of a water sample? **Drop-count titration** TDS meter	

	硬度及鹼度的建議方法		pH meter Test strips 問題編號: **100010842711** 下列哪一個為建議咖啡設備技師用來測量水質樣本的總硬度的測量方式? **<u>滴定法</u>** TDS 測量儀 PH 儀 測量試紙	
			Question ID: 100010842712 Which of the following testing methods is recommended for use by a coffee equipment technician to measure the alkalinity of a water sample? **<u>Drop-count titration</u>** TDS meter pH meter Test strips 問題編號: **100010842712** 下列哪一個為建議咖啡設備技師用來測量水質樣本鹼度的測量方式? **<u>滴定法</u>** TDS 測量儀 PH 儀 測量試紙	

			Question ID: 100010842713 Which **two (2)** of these coffee brewing water parameters are often measured using the drop count titration method? **Total hardness** **Alkalinity** Total Dissolved Solids Temperature 問題編號: **100010842713** 下列哪二個咖啡沖煮參數通常會用滴定法來加以測量? **總硬度** **鹼度** 總溶解固體含量 溫度	
	B. Demonstrate the proper use of drop-count titration to measure the total hardness and alkalinity of a water sample. B.示範正確使用滴定法來測量水質樣本的總硬度和鹼度	Activity 1.06, See Appendix A - Activity Preparation Guide 活動 1.06 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	
1.06.04 **Using a pH Meter** **PH 儀的使用**	A. Recognize that a pH meter is a recommended method for assessing the pH of a water sample B. Describe the calibration process and storage recommendations for a pH meter A. 辨別 PH 儀為測量水質樣本 PH 值的建議方法		**Question ID: 100010842736** Is the following statement true or false? The electrode on a pH meter has a limited-service life and may need to be replaced periodically. **True** False 問題編號: **100010842736** 請問下方陳述正確與否? pH 儀的電極有其有限的使用壽命且可能會需要週期性的	

	B. 描述 PH 儀的建議校正方式及儲存模式		更新 **是** 否	
			Question ID: 100010842737 Which of the following steps are recommended before using a pH meter? **Calibrate using an appropriate buffer solution** Clean thoroughly using alcohol wipes Ensure that the electrode bulb is thoroughly dry Store fully submerged in distilled water 問題編號: **100010842737** 在使用 pH 儀前,下列步驟哪個是建議選項? **用適當的緩衝溶液進行校正** 用酒精徹底擦拭 確定前端電極球徹底乾燥 存放時完整地浸泡於乾淨的蒸餾水中	
			Question ID: 100010842738 How should a pH meter be stored? **With a small amount of storage solution in the cap** Fully submerged in distilled water With a clean and thoroughly dry bulb Upside-down 問題編號: **100010842738** pH 儀應該如何保存呢? **蓋中需有少量的儲存溶液** 整支浸泡於蒸餾水中 有著乾淨且完全乾燥的電極球 上下顛倒	

	C. Demonstrate the proper use of a pH meter to measure the pH of a water sample. C. .示範正確使用 pH 儀來測量水質樣本的 pH 值	Activity 1.06, See Appendix A - Activity Preparation Guide 活動 1.06 看附錄 A 活動準備指南	**This objective has an activity.** **此課程目標有一個活動**	
1.06.05 **Using Water Testing Strips** **使用水質檢測試紙**	A. Recognize that water testing strips are a recommended method for assessing Chlorine, Chlorides, and other parameters of a water sample A. 辨別水質檢測試紙為測量水質樣本中的氯,氯化物及其他水中參數的建議方法		**Question ID: 100010842739** Which of the following testing methods is recommended for use by a coffee equipment technician to measure chlorine levels of a water sample? **Test strips** Drop-count titration TDS meter pH meter 問題編號: **100010842739** 下列哪一個為建議咖啡設備技師用來測量水質樣本中氯含量的方式? **檢測試紙** 滴定法 TDS 測量儀 pH 儀	
			Question ID: 100010842740 Which **two (2)** of the following testing methods may be used by a coffee equipment technician to measure the pH of a water sample? **Test strips** **pH meter** TDS meter Refractometer 下列的測量方式中,哪二個是咖啡設備技師測量水樣本 pH 值的方法?	

			檢測試紙 pH 儀 TDS 測量儀 屈光儀	
			Question ID: 100010842741 Which **two (2)** of these coffee brewing water parameters are often measured using test strips? **Chlorine** **pH** Total Dissolved Solids Temperature 問題編號: **100010842741** 下列哪二個咖啡沖煮水的參數通常用檢測試紙加以測量? **氯** **pH 值** 總溶解固體含量 溫度	
	B. Demonstrate the proper use of test strips to measure the chlorine level of a water sample. B.示範正確使用檢測試紙來測量水質樣本中的氯含量	Activity 1.06, See Appendix A - Activity Preparation Guide 活動 1.06 看附錄 A 活動準備指南	**This objective has an activity.** **此課程目標有一個活動**	
1.06.06 **Using the SCA Water Chart** **使用 SCA 水質圖**	A. Demonstrate the calculation to convert measurements from units of Degrees of hardness or Grains per Gallon to ppm CaCO3	Activity 1.06, See Appendix A - Activity Preparation Guide 活動 1.06 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	

	B. Demonstrate the proper technique for recording total hardness and alkalinity measurements on the SCA Water Chart A. 示範將硬度的測量值由單位德制硬度,克/美式加崙轉換為 PPM 碳酸鈣的計算方式 B. 示範在 SCA 水質圖上紀錄總硬度和鹼度測量值的正確技巧	Activity 1.06, See Appendix A - Activity Preparation Guide 活動 1.06 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	

1.07 | Section | Water Treatment System Maintenance 1.07 | 章節 | 水處理設備系統的保養維修

3 Topic 三個主題

Topic 主題	Objectives 目標	AST Notes 講師附註	Written Online Exam Questions 線上筆試問題	Resources 相關資源
1.07.01 **Water Treatment System Maintenance Requirements** **水處理系統維護的需求**	A. Recognize that water treatment system media have a fixed capacity and limited life span and must be periodically replaced or recharged A. 辨別水處理系統濾芯有一定的使用量限制及有限的壽命,必需週期性地進行更新及置換		**Question ID: 100010843082** Which of the following factors should be considered when determining if a water treatment system cartridge should be replaced? The amount of time the cartridge has been in service The amount of water which the cartridge has treated The amount of dissolved material in the incoming water **All of the above** 問題編號: **100010843082** 下列哪些因素被認為可決定水處理系統的濾芯需要更換?	

			濾芯已使用時間 濾芯已處理的水量 進水水質的礦物質溶解量 **以上皆是**	
			Question ID: 100010843084 What is the likely result of using a water softening cartridge beyond its rated capacity? **Reduced softening performance** Increased softening performance No change in softening performance Increased water flow rate 問題編號: **100010843084** 當水質軟化濾芯超過它的使用量限制,則可能會有什麼樣的結果? **較差的軟化功效** 加強的軟化功效 在水質軟化功效上沒有改變 增加水的流過率	
			Question ID: 100010843088 Is the following statement true or false? Water treatment system cartridges have a limited life span and should be replaced as part of a preventive maintenance plan. **True** False 問題編號: **100010843084** 以下論述正確與否?水處理設備濾芯有其有效的使用壽命且應該以更換作為預防性維護計畫的一部分 **正確** 錯誤	

1.07.02 **Maintaining a Quick-change Cartridge System** **維護快拆濾芯系統**	A. List the common steps for replacing quick-change cartridges, including: 1. Turn off water supply to the system 2. Remove the spent cartridge 3. Install the new cartridge 4. Restore water supply to the system 5. Flush the system as recommended 6. Mark the installed and recommended replacement dates on the cartridge A. 列出一般置換快拆濾芯的步驟,包含: 1. 將系統的水供給截斷 2. 將用過的濾芯拆下 3. 換上新的濾芯 4. 恢復水供給 5. 按建議方式來沖洗系統 6. 將此次濾芯更換日期及下次建議更換的日期紀錄於濾芯上		**Question ID: 100010843090** Which of the following steps is usually recommended before removing a water treatment system cartridge? **Turn off water supply to the system** Flush water through the system Make sure the water supply is on None of the above 問題編號: **100010843090** 在移除水處理設備的濾芯前,下列步驟中哪一個是通常建議的? **將系統的水供給截斷** 以水沖洗整個系統 確認水供給是有的 以上皆非	
			Question ID: 100010843092 Which of the following steps is usually recommended after installing a new carbon filter system cartridge? **Flush water through the system** Turn off water supply to the system Make sure the water supply is on None of the above 問題編號: **100010843092** 在安裝上新的活性碳過濾濾芯後,下列步驟中哪一個是通常建議的? **以水沖洗整個系統** 將系統的水供給截斷 確認水供給是有的 以上皆非	

			Question ID: 100010843093 Is the following statement true or false? All water treatment system heads include a feature which stops water flow through the system when a cartridge is removed. **False** True 問題編號: **100010843093** 以下論述正確與否?所有的水處理系統濾頭都會在濾芯移除後自動停止水流經系統中 **錯誤** 正確	
1.07.03 **Maintaining a Drop-in Cartridge System** **維護置入式過濾濾芯系統**	A. List the common steps for replacing drop-in cartridges, including: 1. Turn off water supply to the system 2. Release system pressure 3. Unscrew the housing from the system head 4. Remove the spent cartridge and o-ring 5. Clean and sanitize the housing 6. Install a new cartridge into the housing 7. Lubricate and install an o-ring into the housing 8. Reassemble housing to the system head 9. Restore water supply to the system 10. Flush the system as recommended 11. Check housing seal for leaks 12. Record the installed and change		**Not tested.** **無測驗**	

	date A. 列出一般更換置入式過濾濾芯的方法,包含: 1. 將系統的水供給截斷 2. 釋出系統壓力 3. 將濾水器外殼自濾頭上鬆開 4. 將用過的濾芯及 O 環移除 5. 清潔並消毒濾水器外殼 6. 將新濾芯裝進濾水器外殼 7. 潤滑並將 O 環放回濾水器外殼 8. 將濾水器外殼鎖回系統濾頭 9. 恢復系統水供給 10. 按建議以水沖洗系統 11. 確認濾水器外殼鎖緊無洩露 12. 紀錄此次濾芯更換日期及下次預計更換日期			
	B. Demonstrate steps for replacing a drop-in cartridge B. 示範更換置入式濾芯的步驟	Activity 1.07.03, See Appendix A - Activity Preparation Guide 活動 1.07.03 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	

1.08 | Section | Preventive Maintenance Components 1.08 | 章節 | 預防性維護零件

5 Topics 五個主題

Topic 主題	Objectives 目標	AST Notes 講師	Written Online Exam Questions 線上考試問題	Resources 相關資源
1.08.01 **Espresso Machine Brewing Group Components** **義式咖啡機沖煮頭零件**	A. Identify common parts of an espresso machine brewing group A. 確認義式咖啡機沖煮頭的一般零件	Activity 1.10, See Appendix A - Activity Preparation Guide	**This objective has an activity.** **此課程目標有一活動**	
	B. List espresso machine brewing group components which are often replaced during a PM service, including: - Group gasket - Group screen - Group screw - Portafilter basket - Portafilter basket retaining spring B. 列出通常由預防性維護服務所更換的義式咖啡機沖煮頭零件,包含: -沖煮頭墊圈 -沖煮頭分水網		**Question ID: 100010843102** Which of the following espresso machine brewing group parts are often replaced during a preventive maintenance visit? Group gasket Group screen Group screw **All of the above** 問題編號: 下方義式咖啡機沖煮頭零件中,哪一個通常會在預防性維護服務的時候更換 沖煮頭墊圈 沖煮頭分水網 分水網螺絲 **以上皆是**	

	-分水網螺絲 -把手濾籃 -把手濾籃固定彈簧 C. List espresso machine brewing group components which are typically cleaned and inspected during a PM service, including: - Group head - Dispersion plate/disc - Locking bezel - Brew valve - Portafilter C. 列出通常由預防性維護服務提供清潔及檢查的義式咖啡機沖煮頭零件 -沖煮頭 -分水塊 -固定框架 -沖煮閥 -沖煮把手		**Question ID: 100010843104** Which of the following espresso machine brewing group parts are often cleaned and inspected during a preventive maintenance visit? Group head Dispersion disc Portafilter **All of the above** 問題編號: **100010843104** 下方義式咖啡機沖煮頭零件中,哪一個通常會在預防性維護服務的時候清潔檢查? 沖煮頭 分水塊 沖煮把手 以上皆是	
			Question ID: 100010843105 Which of the following espresso machine brewing group parts are usually not replaced during a preventive maintenance visit? **Portafilter** Group gasket Group screen Portafilter basket 問題編號: **100010843105** 下方義式咖啡機沖煮頭零件中,哪一個通常不會在預防性	

			維護服務的時候更換 **過濾把手** 沖煮頭墊圈 沖煮頭分水網 分水網螺絲	
1.08.02 **Espresso Machine Steam Valve Components** **義式咖啡機蒸汽閥門零件**	A. Identify common parts of an espresso machine steam valve. A. 辨認義式咖啡機蒸氣閥門的一般零件	Activity 1.10, See Appendix A - Activity Preparation Guide 活動 1.10 看附錄 A 活動準備指南	**This objective has an activity.** 這個課程目標有一個活動	
	B. Identify common parts of a ball-end style espresso machine steam wand joint assembly. B 辨認義式咖啡機的球型接頭蒸氣棒的一般零件	Activity 1.10, See Appendix A - Activity Preparation Guide 活動 1.10 看附錄 A 活動準備指南	**This objective has an activity.** 這個課程目標有一個活動	
	C. List espresso machine steam valve components which are typically replaced during a PM service, including: - seals - o-rings - gaskets - springs - bushings C. 列出通常由預防性維護服務所更換的義式咖啡機蒸氣閥門零件,		**Question ID: 100010843106** Which of the following espresso machine steam valve or wand parts are often replaced during a preventive maintenance visit? **Steam valve seal** Steam valve handle Steam wand Steam wand tip 問題編號: **100010843106** 下列義式咖啡機蒸氣閥或蒸氣棒零件中,哪一個是預防性	

	包含: - 止漏膠墊 - O 形環 - 墊片 - 彈簧 - 套管 D. List espresso machine steam valve components which are typically cleaned and inspected during a PM service, including: - valve bodies - stems - handles - ball ends D. 列出通常由預防性維護服務所檢查及清潔的義式咖啡機蒸氣閥門零件,包含: -蒸氣閥體 -蒸氣閥頂桿 -扳桿 -球頭		維護服務時需更換的? **<u>蒸氣止漏膠墊</u>** 蒸氣扳桿 蒸氣棒 蒸氣噴頭	
			Question ID: 100010843107 Which of the following espresso machine steam valve or wand parts are often cleaned and inspected during a preventive maintenance visit? **<u>Steam valve body</u>** Steam valve seal Steam valve o-rings Steam wand o-rings 下列義式咖啡機蒸氣閥或蒸氣棒零件中,哪一個是預防性維護服務時需進行清潔檢查的? **<u>蒸氣閥體</u>** 蒸氣止漏膠墊 蒸氣閥中的 O 型環 蒸氣棒中的 O 型環	
			Question ID: 100010843108 Which of the following espresso machine steam valve or wand parts are often replaced during a preventive maintenance visit? Steam valve seal Steam valve o-rings Steam wand o-rings	

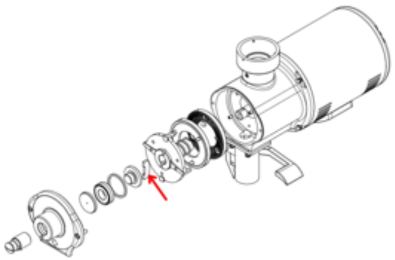
			All of the above 問題編號: **100010843108** 下列義式咖啡機蒸氣閥或蒸氣棒零件中,哪一個是預防性維護服務時需更換的? 蒸氣止漏膠墊 蒸氣閥中的 O 型環 蒸氣棒中的 O 型環 **以上皆是**	
1.08.03 **Espresso Machine Steam Boiler Components** **義式咖啡機的蒸氣鍋爐零件**	A. Identify common parts of an espresso machine steam boiler which are typically inspected or maintained during a PM service A. 辦認通常由預防性維護服務檢查或維護的義式咖啡機蒸氣鍋爐一般零件.	Activity 1.10, See Appendix A - Activity Preparation Guide 活動 1.10 看附錄 A 活動準備指南	**This objective has an activity.** 這個課程目標有一個活動	
	B. List common parts of an espresso machine steam boiler which are typically inspected or maintained during a PM service, including: - Pressure safety valve - Anti-vacuum valve - Water level probe - Pressure switch B. 列出通常由預防性維護服務檢查或維護的義式咖啡機蒸氣鍋爐		**Question ID: 100010843110** Which of the following espresso machine steam boiler parts are often inspected during a preventive maintenance visit? Pressure safety valve Anti-vacuum valve Water level probe **All of the above** 問題編號: **100010843110** 下列義式咖啡機鍋爐零件中,哪一個是預防性維護服務時需檢查的?	

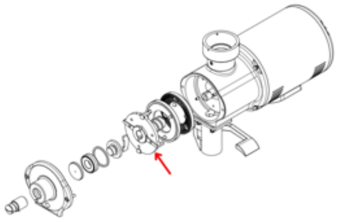
	一般零件, 包含: -壓力安全閥 -真空排除閥 -水位探針 -蒸氣壓力接觸開關		壓力安全閥 真空排除閥 水位探針 **以上皆是**	
			Question ID: 100010843111 When inspecting an espresso machine anti-vacuum valve, which **two (2)** of the following signs indicate that the component is working properly? **The valve does not leak steam when sealed** **The valve allows some steam to escape when the stem is depressed** The valve allows some steam to escape when sealed The valve remains sealed when the stem is depressed 問題編號: **100010843111** 當檢查義式咖啡機的真空排除閥時,下面哪二個狀態可以判斷零件正常運作? **當鎖緊時閥門並無洩露** **當蒸氣閥頂桿壓下時,閥門可讓一些蒸氣散逸而出** 當鎖緊時閥門可讓一些蒸氣散逸而出 當蒸氣閥頂桿壓下時,閥門仍然鎖緊	

			Question ID: 100010843112 When inspecting an espresso machine pressure safety valve, which **two (2)** of the following signs indicate that the component should be replaced? **The valve allows some steam to escape at normal boiler operating pressure** **The manufacturer's tamper-resistant feature has been removed or damaged** The valve remains sealed at normal boiler operating pressure The valve allows steam to escape at the rated release pressure 問題編號: **100010843112** 當檢查義式咖啡機的安全閥時,下面哪**二個**狀態可以判斷零件需要更換? **在正常鍋爐運作壓力下,閥門可以讓一些壓力散逸而出** **製造商在安全閥上的印記標章被移除或破壞** 在正常鍋爐運作壓力下閥門完全緊閉 蒸氣從閥門以洩壓速度散逸而出	
1.08.04 **Batch Brewer Components**	A. Identify common parts of a batch coffee brewer which are typically inspected or maintained during a PM service.	Activity 1.10, See Appendix A - Activity Preparation Guide 活動 1.10 看附錄 A 活動準備指南	**This objective has an activity.** 這個課程目標有一個活動	

滴濾式咖啡機零件	B. List common parts of a batch coffee brewer which are typically inspected or maintained during a PM service, including: - Water level probe - Spray head A. 辨認通常由預防性維護服務檢查或維護的滴濾式咖啡機一般零件,包含: -水位探針 -噴灑頭		**Question ID: 100010843118** Which of the following batch coffee brewer parts are typically inspected and cleaned during a preventive maintenance visit? **Water level probe** Heating element Brew dispensing valve Solid state relay 問題編號: **100010843118** 下列滴濾式咖啡機零件中,哪一個是預防性維護服務時需清潔檢查的? **水位探針** 加熱零件 沖煮分配閥 固態繼電器	
			Question ID: 100010843119 Which **two (2)** of the following batch coffee brewer parts are typically inspected and cleaned during a preventive maintenance visit? **Spray head** **Water level probe** Heating element Auto fill valve 問題編號: **100010843119** 下列滴濾式咖啡機零件中,哪二個是預防性維護服務時需	

			清潔檢查的? **噴灑頭** **水位探針** 加熱零件 自動補水閥	
			Question ID: 100010843120 Is the following statement true or false? Batch coffee brewers are simple devices which do not require preventive maintenance. **False** True 問題編號: **100010843120** 以下論述正確與否?批量咖啡沖煮機是簡單的裝置,無需預防性維護. **錯誤** 正確	
1.08.05 **Coffee Grinder Components** **咖啡磨豆機的零件**	A. Identify common parts of coffee grinder which are typically inspected or maintained during a PM service. A. 確認通常由預防性維護服務檢查或維護的磨豆機一般零件	Activity 1.10, See Appendix A - Activity Preparation Guide 活動 1.10 看附錄 A 活動準備指南	**This objective has an activity.** 這個課程目標有一個活動	

	B. List common parts of coffee grinder which are typically inspected or maintained during a PM service, including: - Bean hopper - Grinding chamber - Grinding burrs - Burr carrier - Burr carrier shear plate* - Outlet chute - Declumping device* B. 列出通常由預防性維護服務檢查或維護的磨豆機一般零件,包含 -咖啡豆槽 -研磨腔室 -咖啡刀盤 -刀座 -刀座插銷 -出粉通道 -防結塊裝置	*Note that grinder designs vary by manufacturer and not all features are found on all grinders *請注意磨豆機的設計因製造商而異,並非所有的特點都可以在所有的磨豆機上找到	**Question ID: 100010843121** Which of the following espresso grinder parts are often inspected and cleaned during a preventive maintenance visit? Grinding burrs De-clumping device Bean hopper **All of the above** 問題編號: **100010843121** 下列咖啡磨豆機零件中,哪一個是預防性維護服務時需清潔檢查的? 磨豆機刀盤 防結塊裝置 咖啡豆槽 以上皆是	[Ersatzteilliste Spare parts list EK43_EK43S DE_EN_V2.0.3.indd (shopify.com)](shopify.com)
			Question ID: 100010843122 Which coffee grinder part is indicated by the arrow in this parts diagram?  **Burr carrier shear plate** Burr carrier Grinding burr	

			Grinding chamber 問題編號: **100010843122** 圖中箭頭所指是咖啡磨豆機的哪一個部分? **<u>刀盤軸心插銷</u>** 刀盤 磨刀 研磨室	
			Question ID: 100010843123 Which coffee grinder part is indicated by the arrow in this parts diagram?  **<u>Burr carrier</u>** Grinding burr Bean hopper Grinding chamber 問題編號: **100010843123** 圖中箭頭所指是咖啡磨豆機的哪一個部分? **<u>刀座</u>** 刀盤 咖啡豆槽 咖啡粉槽	

1.09 | Section | Maintenance Strategies 1.09 | 章節 | 保養策略

2 Topics 二個主題

Topic 主題	Objectives 目標	AST Notes 講師附註	Written Online Exam Questions 線上考試問題	Resources 相關資源
1.09.01 **Preventive versus Reactive Maintenance** **預防式與回應式的維護**	A. Define preventive maintenance as regular and routine service performed to maintain proper system function and reduce unexpected equipment failure. B. Define reactive maintenance as unplanned repair of equipment that has failed in order to restore proper system function. A. 定義預防式維護為定期常規的服務執行以維持恰當的系統功能並減少無法預期的設備故障 B. 定義回應式維護為設備故障後,為了讓系統回到正常功能運作而進行未在計畫內的修復	Though a good preventive maintenance program should reduce the likelihood of unexpected equipment failure, no maintenance program can completely eliminate this possibility. Most programs seek to balance program cost vs risk of downtime, a topic which will be explored in higher-level courses. 因此,一個良好的預防性維護計畫應該會減少無法預期的設備故障發生的可能性,沒有任何的維護計畫可以完全的消除未預期故障發生的可能性. 多數的計畫尋求花費成本與停機時間風險的平衡,這個課題我們會在更高階課程中再進一步的探討	**Question ID: 100010843124** Which of the following best describes the purpose of coffee equipment preventive maintenance? **Regular and routine service to maintain proper system function** Unplanned repair of equipment that has already failed Regular service in order to eliminate all possibility of unexpected equipment failure Unnecessary service scheduled in order to increase technician revenue 問題編號: **100010843124** 下列何者是咖啡設備預防性維護目的的最佳描述? **定期常規的服務執行以維持恰當的系統功能** 未在計畫內的修理設備的故障 為了減少所有未預期的設備故障而執行的定期服務 為了增加技師的收入而計畫不需要的服務	

			Question ID: 100010843125 Which of the following best describes coffee equipment reactive maintenance? **Unplanned repair of equipment that has failed in order to restore proper system function** Regular and routine service to maintain proper system function Maintenance recommended by the equipment manufacturer according to expected component lifespan Service to replace components which show signs of manufacturing defects 問題編號: **100010843125** 下列何者是咖啡設備回應性維護的最佳描述? **設備故障後,為了讓系統回到正常功能運作而進行未在計畫內的修復** 定期常規的服務執行以維持恰當的系統功能 設備製造商因應零件的壽命期限而建議的維護 提供服務以更換有製造瑕玼徵兆的零件	
			Question ID: 100010843126 Which of the following best describes an espresso machine service which is planned to be performed every six months to rebuild brewing groups and steam valves? **Preventive maintenance** Reactive maintenance Installation service Warranty service	

			問題編號: **100010843126** 下列何者絕佳的形容一家義式咖啡機設備商計畫每六個月提供對沖煮頭及蒸氣閥門的維護? **預防性維護** 回應性維護 義大利式維護 保修服務	
1.09.02 Recommended Maintenance Intervals 建議的保養週期	A. Recognize that different components will have different recommended maintenance intervals and expected service lives A. 認知到不同的零件有不同建議的保養週期及可預期的使用壽命 B. List factors which impact the longevity of components,including: - wear due to use and abuse - degradation due to high-temperature operating environment - degradation due to duty cycling or extended operation - loss of function due to scale buildup B. 列出會影響零件壽命的因素 -因使用及濫用而引起的耗損 -因高溫的作業環境而引起的損害 -因工作周期或延長的作業而引起的		**Question ID: 100010843127** Which of the following factors may affect the expected life span of coffee equipment components? Wear due to normal use High temperature operating environment Buildup of limescale **All of the above** 問題編號: **100010843127** 下列哪一個因素可能會影響咖啡設備零件的壽命期限? 一般使用所引起的損耗 高溫作業環境 水垢的累積 **以上皆是** **Question ID: 100010843128** Is the following statement true or false? Preventive maintenance is not necessary for an espresso machine which is powered on all day, every day, but not used very often. **False** True	

	損害 -因水垢而引起的功能喪失 C. Recognize that equipment manufacturers may recommend preventive maintenance schedules for their equipment C. 認知設備製造商可能會建議設備所需的預防性維護週期		問題編號: **100010843128** 以下的論述正確與否?每天且一整天都打開卻不常用的義式咖啡機是不需要預防性維護的. **錯誤** 正確 **Question ID: 100010843129** Which **two (2)** statements best describe recommended preventive maintenance for two identical espresso grinders, one which grinds 10kg of coffee per week and the other which grinds 2kg per week? **Regular preventive maintenance for both grinders** **Different grinder burr replacement schedules for each grinder** Regular preventive maintenance for one grinder only The same grinder burr replacement schedule for both grinder 問題編號: **100010843129** 二台相同義式磨豆機,一台每週磨 10kg 咖啡,另一台每週磨 2kg 咖啡,下列哪二個論述最好的描述了它們的預防性維護? **二台磨豆機都需要進行週期性的預防性維護** **每台磨豆機有不同的刀盤更換週期** 只有一台磨豆機需要週期性的預防性維護 二台磨豆機都適用一樣的刀盤更換週期	

1.10 | Section | Preventive Maintenance Tools and Supplies 1.10 | 章節 | 預防性維護的工具與零件消耗品

2 Topics 二個主題

Topic 主題	Objectives 目標	AST Notes 講師附註	Written Online Exam Questions 線上筆試問題	Resources 相關資源
1.10.01 Recommended Hand Tools 建議的手動工具	A. List common hand tools used for maintaining espresso machine brewing group, including: - Pick awl - Short screwdriver - Stiff bristle cleaning brush - Blank portafilter basket A. 列出一般維護義式咖啡機沖煮頭時所需手動工具, 包含: - 尖錐 - 短柄螺絲起子 - 硬毛清潔刷 - 無孔濾杯 B. List common hand tools used for maintaining espresso machine steam valve, including: - Wrenches - Pick awl - Stiff bristle cleaning brush - Snap ring pliers B. 列出一般維護義式咖啡機蒸氣閥時所需手動工具, 包含:		**Question ID: 100010844140** Which **two (2)** of the following tools are often recommended for replacing an espresso machine group gasket? **Pick awl** **Stiff-bristle cleaning brush** Wrench Snap ring pliers 問題編號: **100010844140** 下列哪二個工具通常建議在更換義式咖啡機沖煮頭墊圈時使用? **尖錐** **硬毛刷** 扳手 卡環鉗	
			Question ID: 100010844141 Which of the following tools is often recommended for replacing coffee grinder burrs? **Screwdriver** Pick awl Snap ring pliers Wrench	

	- 扳手 - 尖錐 - 硬毛刷 - 卡環鉗 C. List common hand tools used for maintaining batch brewers, including: - Screwdrivers - Wrenches - Stiff bristle cleaning brush - Large water pitcher or container C. 列出一般維護滴濾式咖啡機時所需手動工具, 包含: -螺絲起子 -扳手 - 硬毛刷 - 大水瓶或水桶 D. List common hand tools used for maintaining a coffee grinder, including: - Screwdrivers - Soft bristle brush - Stiff bristle cleaning brush - Vacuum cleaner D. 列出一般維護義式咖啡磨豆機時所需手動工具, 包含: - 螺絲起子 - 軟毛刷		問題編號: **100010844141** 下列哪個工具通常建議在更換義式咖啡磨豆機磨盤時使用? **螺絲起子** 尖錐 卡環鉗 扳手 **Question ID: 100010844142** Which of the following tools is often recommended for servicing a ball-joint style steam wand? **Wrench** Pick awl Screwdriver Snap ring pliers 問題編號: **100010844142** 下列哪個工具通常建議在更換義式咖啡機球型接頭蒸氣棒時使用? **扳手** 尖錐 螺絲起子 卡環鉗	

	- 硬毛刷 - 吸塵器			
1.10.02 Recommended Consumables **建議消耗品**	A. List the recommended supplies used to clean an espresso machine and batch brewer, including: - Espresso machine detergent - Coffee brewer detergent - Milk system cleaning solution B. List the types of lubricants which may be recommended for maintaining espresso machines and coffee grinders, including: - Silicone o-ring lubricant - Grinder collar thread grease* A. 列出一般建議用於清潔一台義式咖啡機及滴濾式咖啡機所需消耗品,包含: - 義式咖啡清潔劑 - 滴濾式咖啡機清潔劑 - 牛奶系統的清潔溶劑 B. 列出可能適用於維護義式咖啡機及義式咖啡磨豆機的潤滑劑類型,包含:	Examples of o-ring lubricant include Dow 111/Molykote and Examples of grinder collar thread grease include Kluberlub NH1 11/222 舉例 O 環潤滑油包含道康寧 Dow 111/Molykote,上刀座固定螺紋潤滑油包含克魯勃 Kluberlub NH1 11/222。	**Question ID: 100010844197** Which of the following cleaning supplies is recommended when servicing an espresso machine brewing group? **Espresso machine detergent** Milk system cleaner Kitchen equipment degreaser WD-40 問題編號: **100010844197** 下列哪一個清潔消耗品建議用於維護義式咖啡機沖煮頭? **義式咖啡機清潔劑** 牛奶系統清潔劑 廚房設備除油劑 WD-40 防鏽潤滑劑	
			Question ID: 100010844247 Which of the following characteristics should lubricants used with espresso machine o-rings have? Suitable for high-temperature applications Approved for incidental food contact Suitable for contact with water **All of the above**	

	-矽 O 環潤滑油 -上刀座固定螺紋潤滑油		問題編號: **100010844247** 下列哪一個特性是用於義式咖啡機 O 型環潤滑油應具備的? 適用於高溫的應用 核準可偶發性的與食物接觸 適於與水接觸 **以上皆是**	
			Question ID: 100010844248 Is the following statement true or false? Lubricant used on a threaded espresso grinder adjustment collar should be approved for incidental food contact. **True** False 問題編號: **100010844248** 下列論述正確與否? 使用於義式磨豆機上刀座固定螺紋的潤滑劑需核準可偶發性的與食物接觸 **正確** 錯誤	

1.11 | Section | Performing Maintenance 1.11 | 章節 | 維護的實施

7 Topics 七個主題

Topic 主題	Objectives 目標	AST Notes 講師附註	Written Online Exam Questions 線上筆試問題	Resources 相關資源
1.11.01 **Safety During a Preventive Maintenance Service** **預防性維護服務過程的安全**	A. State that, to minimize potential injury, coffee equipment must be powered off and unplugged whenever maintenance requires that panels be removed, or grinders be disassembled A. 指出為了降低潛在傷害, 不論維護需要拿掉隔板或拆開磨豆機,咖啡設備必需在關閉電源且拔掉插頭下進行		**Question ID: 100010844257** Which of the following espresso machine preventive maintenance activities require the equipment to be powered off and unplugged? **Inspecting pressure switch contacts** Cleaning an espresso machine group Replacing espresso machine group gaskets Adjusting pump pressure 問題編號: **100010844257** 下列哪一個義式咖啡機的預防性維護活動需要將設備關閉電源並拔掉插頭? **檢查壓力接觸開關** 清潔義式咖啡機沖煮頭 更換義式咖啡機沖煮頭墊圈 調整泵浦壓力	

			Question ID: 100010844259 Which of the following batch brewer preventive maintenance activities require the equipment to be powered off and unplugged? **Removing and cleaning level probe** Inspecting hot water tap function Removing and cleaning spray head Confirming brew volumes 問題編號: **100010844259** 下列哪一個滴濾式咖啡機的預防性維護活動需要將設備關閉電源並拔掉插頭? **移除並清潔水位探針** 檢查熱水出水功能 移除並清潔噴灑頭 確認沖煮水量	
			Question ID: 100010844260 Which of the following coffee grinder preventive maintenance activities require the equipment to be powered off and unplugged? **Replacing grinder burrs** Adjusting grinder setting Calibrating grind adjustment knob Adjusting programmed grinding time 問題編號: **100010844260** 下列哪一個咖啡磨豆機的預防性維護活動需要將設備關閉電源並拔掉插頭?	

			更換研磨刀盤 調整研磨設定 校正研磨刻度調整旋鈕 調整計畫性的研磨時間	
1.11.02 **Preventive Maintenance Checklist** **預防性維護檢查表**	A. Recognize that a preventive maintenance checklist is recommended to ensure that no maintenance steps are missed and provide a record of services performed A. 認知預防性維護檢查表建議用來確保沒有任可一個維護步驟會被跳過且提供服務實施的紀錄		**Not tested.** **無測驗**	
	B. Demonstrate the use of a preventive maintenance checklist to record preventive maintenance service activities. B. 示範預防性維護檢查表的使用,以紀錄預防性維護服務活動	Activity 1.13.02, See Appendix A - Activity Preparation Guide 活動 1.13.02 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	
1.11.03 **Espresso Machine Preventive Maintenance Cleaning Steps** **義式咖啡機預防性維護**	A. List common cleaning steps which are recommended during an espresso machine preventive maintenance service, including: - Cleaning the brewing group and dispersion disc - Cleaning portafilter - Detergent backflush the brewing		**Question ID: 100010844421** Which of the following cleaning activities are recommended as part of an espresso machine preventive maintenance service? Cleaning the brewing group Detergent backflush the brewing group Cleaning steam tip	

清潔步驟	group - Cleaning steam wand and tip A. 列出義式咖啡機預防性維護服務時,一般建議進行的清潔步驟, 包含: -清潔沖煮頭及分水塊 -清潔沖煮把手 -以清潔劑逆洗沖煮頭 -清潔蒸氣棒及噴頭		**All of the above** 問題編號: **100010844421** 下列何者的清潔活動建議為義式咖啡機預防性維護服務的一部分? 清潔沖煮頭 以清潔劑逆洗沖煮頭 清潔蒸氣棒及噴頭 以上皆是	
			Question ID: 100010844429 Which of the following cleaning activities are **NOT** usually recommended as part of an espresso machine preventive maintenance service? **Disassemble and clean brew valve** Cleaning the brewing group Cleaning the portafilter Detergent backflush the brewing group 問題編號: **100010844429** 下列何者的清潔活動通常不建議為義式咖啡機預防性維護服務的一部分? **拆開並清潔沖煮閥** 清潔沖煮頭 清潔沖煮把手 以清潔劑逆洗沖煮頭	

			Question ID: 100010844433 From the drop-down menu, choose the correct step to fill in the blank to reflect recommended espresso machine brewing group backflushing procedure: 1. Insert blank basket or cleaning disk into portafilter 2. _ 3. Insert portafilter into group 4. _ 5. Remove portafilter and rinse group and portafilter 6. _ 7. Activate brew cycle for 10 seconds, stop 10 seconds, repeat 5 times 8. Replace standard basket in portafilter or remove backflush disk A = Add recommended amount of espresso detergent B = Activate brew cycle for 10 seconds, stop 10 seconds, repeat 5 times C = Insert portafilter without detergent into group **A = 2 / B = 4 / C = 6** 問題編號: **100010844433** 由下拉式選單中,選取正確的步驟填入空白,以反應正確的義式咖啡機沖煮頭逆洗流程 1. 將無孔濾杯或逆洗片置入沖煮把手中	

			2. ________ 3. 將沖煮把手卡入沖煮頭 4. ________ 5. 移開把手並沖洗沖煮頭及沖煮把手 6. _________ 7. 啟動 10 秒的沖煮,然後停止 10 秒,這樣的循環持續五次 8. 將沖煮把手中的濾杯更換回原來的標準濾杯或移除逆洗片 A=加入建議適量的義式咖啡清潔劑 B=啟動 10 秒的沖煮,然後停止 10 秒,這樣的循環持續五次 C=將沒有清潔劑的沖煮把手卡入沖煮頭 A = Add recommended amount of espresso detergent B = Activate brew cycle for 10 seconds, stop 10 seconds, repeat 5 times C = Insert portafilter without detergent into group	
	B. List the steps to backflush an espresso machine brewing group B. 列出義式咖啡機沖煮頭的逆洗步驟	Activity 1.13.03, See Appendix A - Activity Preparation Guide 活動 1.13.03 看附錄 A 活動準備指南	**This objective has an activity.** 這個課程目標有一個活動	
1.11.04 **Espresso Machine Preventive Maintenance Parts Replacement**	A. Demonstrate espresso machine group screen and gasket replacement	Activity 1.13.04-1.13.05, See Appendix A - Activity Preparation Guide 活動 1.13.04-1.13.05 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	

義式咖啡機預防性維護-零件更換	B. Demonstrate espresso machine steam wand joint rebuild A. 示範義式咖啡機沖煮頭的分水網及墊圈的更換方式 B. 示範義式咖啡機蒸氣棒接頭的重新組裝	Activity 1.13.04-1.13.05, See Appendix A - Activity Preparation Guide 活動 1.13.04-1.13.05 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	
1.11.05 **Espresso Machine Preventive Maintenance Inspection Steps** **義式咖啡機預防性維護檢查步驟**	A. List common inspection steps which are recommended during an espresso machine preventive maintenance service, including: - Overall steam leaks - Safety valve steam leaks - Steam valve leaks - Overall water leaks - Brew valve leaks - Anti-vacuum valve function - Pressure switch contact condition - Drain system function - Cold water expansion valve function A. 列出一般義式咖啡機預防性維護服務所建議的檢查步驟,包含: - 所有蒸氣的洩出 - 安全閥門蒸氣的洩出 - 蒸氣閥的洩出	If inspection shows that these operating conditions are not within desired range, adjustment may be required to correct them. Adjustment is covered in higher-level course 如果檢查顯示這些運作的狀況不在預期的範圍內,可能會需會調整來加以修正 調整的內容將在更高級數的課程中	**Question ID: 100010844437** Which of the following espresso machine components should be inspected for possible steam leaks during a preventive maintenance service? Safety valve Steam valve Anti-vacuum valve **All of the above** 問題編號: 在預防性維護服務中,為了蒸氣洩露的可能性,需要檢查下列哪一個義式咖啡機零件? 安全閥 蒸氣閥 真空排除閥 以上皆是	

	- 所有水的排出 - 沖煮閥的排水 - 真空排除閥的功能 - 壓力接觸開關的狀況 - 排水系統功能 -冷水膨脹閥的功能 B. List common espresso machine operating conditions which should be recorded during a preventive maintenance service, including: - Pump pressure - Steam boiler pressure - Group water temperature - Brew volumes (auto-volumetric machines) - Group water debit B. 列出在預防性維護服務中,應該被紀錄下的一般義式咖啡機操作狀況,包含: - 泵浦壓力 - 蒸氣鍋爐壓力 - 沖煮頭的水壓 - 沖煮水量(自動定量咖啡機)		**Question ID: 100010844439** When inspecting the espresso machine for water leaks, for which of the following components is dripping a small amount of water acceptable? **Cold water expansion valve** Brew valve Steam valve Steam joint 問題編號: **100010844439** 當檢查義式咖啡機的排水時,哪一個零件滴出小小量的水是可以接受的? **冷水膨脹閥** 沖煮閥 蒸氣閥 蒸氣棒接頭	
			Question ID: 100010844441 Which of the following espresso machine operating conditions should be inspected during preventive maintenance service? Pump pressure Steam boiler pressure Group water debit **All of the above** 問題編號: **100010844441** 下列哪一個義式咖啡機的運作狀況在預防性維護服務時需要檢查?	

			泵浦壓力 蒸氣鍋爐壓力 沖煮頭的出水量 以上皆是	
	C. Demonstrate the inspection of an espresso machine during preventive maintenance service C. 示範預防性維護服務中所進行的義式咖啡機檢查	Activity 1.13.04-1.13.05, See Appendix A - Activity Preparation Guide 活動 1.13.04-1.13.05 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	
1.11.06 **Batch Brew Preventive Maintenance** **滴濾式咖啡機的預防性維護**	A. List common maintenance steps which are recommended during a batch brewer preventive maintenance service, including: - Inspect and clean water level probe - Inspect and clean spray head - Inspect hot water tap function - Confirm brew volumes A. 列出一般滴濾式咖啡機進行預防性維護服務的建議維護步驟, 包含 -檢查並清潔水位探針 -檢查並清潔噴灑頭 -檢查熱水出水功能 -確認沖煮水量		**Question ID: 100010844445** Which of the following maintenance steps are **NOT** generally recommended during a batch brewer preventive maintenance service? Drain and refill the water tank Inspect and clean water level probe Inspect and clean spray head Confirm brew volumes 問題編號: **100010844445** 在滴濾式咖啡機的預防性維護服務中,下列哪一個維護步驟通常並不建議? **將熱水槽排乾然後重新裝水** 檢查並清潔水位探針 檢查並清潔噴灑頭 確認沖煮水量	

			Question ID: 100010844447 Which of the following maintenance steps is recommended for a coffee batch brewer water level probe during preventive maintenance service? **Inspect and clean to remove buildup** Trim at least 1 cm from the end Replace it Check its capacitance with a multimeter 問題編號: **100010844447** 在滴濾式咖啡機的預防性維護服務中,下列哪一個維護步驟是建議對水位探針進行的? **檢查並清潔以去除水垢** 從尾段修剪 1cm 更換它 用萬用表檢查它的電容	
			Question ID: 100010844449 Which of the following maintenance steps is **NOT** recommended for a coffee batch brewer spray head during preventive maintenance service? **Use a caliper to measure all holes against factory specifications** Inspect and replace if damaged Clean to remove coffee residue Remove any limescale buildup 問題編號: **100010844449** 在滴濾式咖啡機的預防性維護服務中,下列哪一個維護步驟	

			不建議對噴灑頭進行? 使用測徑儀來測量所有的洞與工廠規格作參照 如果毀損需進行檢查且更換 清潔並去除所有的咖啡殘留物 去除所有的水垢	
	B. Demonstrate inspection and cleaning of a batch brewer level probe B. 示範檢查並清潔滴濾式咖啡機的水位探針	Activity 1.13.06, See Appendix A - Activity Preparation Guide 活動 1.13.06 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	
1.11.07 **Espresso Grinder Preventive Maintenance** **義式磨豆機的預防性維護**	A. List common maintenance steps which are recommended during espresso grinder preventive maintenance service, including: - Clean bean hopper - Clean grind chamber throat - Clean and inspect grinder burrs - Replace grinder burrs as-needed - Clean inside grinding chamber - Clean and re-lubricate adjustment collar threads - Calibrate and adjust grind setting per customer specification - Adjust grinding time per customer specification (on-demand grinders only) A. 列出一般義式咖啡磨豆機的預防		**Not tested.** **無測驗**	

	性維護服務建議的維護步驟, 包含: -清潔咖啡豆槽 -清潔研磨腔室頸 -清潔並檢查研磨刀盤 -如果需要更換刀盤 -清潔研磨腔室的內部 -上刀座固定螺紋的清潔並重新上潤滑油 -校正並根據客戶的需求調整研磨設定 -根據客戶的需求調整研磨時間 (只有定量磨豆機)			
	B. Demonstrate replacement of espresso grinder burrs. B. 示範更換義式咖啡機刀盤	Activity 1.13.07, See Appendix A - Activity Preparation Guide 活動 1.13.07 看附錄 A 活動準備指南	**Tested on Practical Exam.** **於實務操作考試中進行測驗**	http://web.archive.org/web/20210620214141/https://www.espressoparts.com/pages/changing-grinder-burrs-when-and-how

5. Essential SCA Training Documents 必要的 SCA 訓練文件

- The SCA Water Chart
- SCA 水質圖

6. Bibliography 參考書目

- Wellinger, Marco, Samo Smrke, Chahan Yeretzian, *The SCA Water Quality Handbook*, Specialty Coffee Association, 2018
- Billemon, Ronny, *Equilibrium in Water and Coffee*, 2018
- Colonna-Dashwood, Maxwell, Christopher H. Hendon, *Water for Coffee*, Maxwell Colonna-Dashwood and Christopher H. Hendon, 2015

7. Required Equipment and Supplies List 所需設備及用品清單

The chart below represents the minimum pieces of equipment and supplies for the instructional part of this course. This does not include instructional supplies like clipboards, aprons, and similar. Please refer to the Water & preventive Maintenance Foundation Practical Exam Trainer Guide for additional items required to conduct the practical exam.

IMPORTANT: Be sure to have an easily accessible, fully-equipped first aid kit on site, along with the appropriate personal protection equipment necessary for the activities of the course and exam.

底下的表格代表此課程在教學部分所需設備及零件的最小量. 這並不包含教學需要的板夾,圍裙及類似的項目. 其他進行考試所需要的額外項目,請參考水&預防性維護基礎實務測驗講師指南,

重要:務必確保現場備有完整配備且方便取用的急救箱,還有課程活動及考試中需要的個人適當的防護配備

Item 品項	Qty 數量	Notes 備註
Coffee Equipment 咖啡設備		
Espresso machine 義式咖啡機	1	Espresso machine must be connected to water supply and electricity 義式咖啡機必需接上水源及電源
Batch Coffee Brewer 滴濾式咖啡機	1	Batch brewer need not be connected to water supply and electricity 滴濾式咖啡機必需接上水源及電源
Espresso Grinder 義式咖啡磨豆機	1	Espresso grinder must be able to be connected to electricity 義式咖啡磨豆機必需接上水源及電源
Espresso Machine Components 義式咖啡機零件		Unless noted, all component quantities are one per (3) three students. In each case, there should be a used/damaged and new component that fits the demonstration espresso machine. 除非另有說明,所有零件數量是三個學生一組. 在每一個項目中,都需要準備與示範義式咖啡機相稱的使用過的/損壞的及全新的零件

Group gaskets 沖煮頭墊圈	1	Have an example of a previously-used and hardened group gasket as well as new gaskets 準備之前使用的墊圈, 已變硬的墊圈及新的墊圈
Group screen 沖煮頭分水網	1	Have an example of a previously-used screen which is deformed or has coffee build-up, as well as new screens 準備之前使用過已變形,使用過仍良好的分水網,及新的分水網
Group screw 分水網螺絲	1	Have an example of a previously-used and damaged screw as well as new 準備之前使用的螺絲,損壞的螺絲及新的螺絲
Portafilter basket 沖煮把手濾杯	1	Have an example of a previously-used and damaged basket as well as new baskets 需準備之前使用的濾杯, 損壞的濾杯及全新濾杯
Portafilter basket retaining spring 沖煮把濾杯固定彈簧	1	Spring must fit portafilter that will be maintained during class activities 彈簧必需與課程活動中維護的把手合適相稱
Steam valve rebuild kit 蒸氣閥重置組	1	Kit should include common replacement parts to rebuild the steam valve used in class activities 組合需包含一般替換零件以便在課程活動中重新組裝蒸氣閥使用
Steam wand joint rebuild kit 蒸氣棒接頭重置組	1	Kit should include common replacement parts to rebuild the steam wand joint used in class activities 組合需包含一般替換零件以便在課程活動中重新組裝蒸氣棒接頭使用
Pressure switch 壓力開關	1	One per class, 30A standard style, may be new or used 一個班一個,30A 標準式,可以是全新或使用過的
Espresso Grinder Components **義式磨豆機零件**		Unless noted, all component quantities are one per three (3) students. 除非另有說明,所有零件的數量都是三個學生一組
Grinding burrs with screws 磨豆機刀盤帶螺絲	1	Display one example each of burrs in good condition and heavily worn burrs, both must be the replacement part for the espresso grinder to be maintained during course activities. 展示樣品需包含一個良好的刀盤，及一個已嚴重磨損的刀盤。在課程活動中，二者都必須是待維護義式磨豆機的替換零件。
Hand and Measurement Tools **手動及量測工具**		Unless noted, all component quantities are listed per three (3) students. 除非另有說明,所有零件的數量都是三個學生一組
Straight screwdrivers 一字螺絲起子	1	Provide assorted sizes, including short bladed 提供各種尺寸,包含短柄的

Phillips head screwdriver 十字螺絲起子	1	Provide assorted sizes 提供各種尺寸
Pick awl 尖錐	1	Provide assorted styles, including straight and angled 提供各種尺寸,包含直的和有角度的
Stiff-bristled cleaning brush 硬毛刷	1	
Blank portafilter basket 無孔濾杯	1	
Adjustable wrench 活動扳手	1	
Set of Combination Wrenches 一組的多用扳手	1	Metric or standard in a range that fits equipment used in the course 課程中所使用的為標準或公制扳手，並在一個範圍內與設備相符合
Snap ring pliers 卡環鉗	1	Either one reversible-style or one each of the internal and external style may be used 可為一支反轉式卡環鉗或內部及外部卡環鉗各一
Soft bristle brush 軟毛刷	1	
Small vacuum cleaner 小型吸塵器	1	
Electrical conductivity/TDS meter 導電度計/TDS 測量儀	1	Handheld version 手持式版本
Handheld pH meter 手持式 pH 測量儀	1	
Titration measurement kit for total hardness/alkalinity		This is a drop count kit. Note that this kit is often labeled as measuring carbonate hardness/KH. 這是一個滴定組. 注意滴定組通常標記為測量 碳酸鹽硬度/KH

總硬度/鹼度滴定測量組		
Consumable Materials **消耗性材料**		Provide sufficient quantities of consumables for all activities and Practical Exam sections 提供足夠的消耗量給所有的活動和實務考試
Espresso machine detergent 義式咖啡機清潔劑		
Coffee brewer detergent 滴濾式咖啡機清潔劑		
Milk system cleaning solution 牛奶系統清潔溶劑		
Silicone o-ring lubricant (NSF) 矽 O 環潤滑油(NSF)		
Grinder collar thread grease (NSF) 上刀座固定螺紋潤滑油(NSF)		
Water test strips for Chloride testing 氯測試的水測驗試紙		
Personal Protective Equipment **個人保護設備**		Unless noted, all component quantities are one per student. 除非另有說明,所有的零件數量皆為一個學生一組
Insulating work gloves 絕緣工作手套	1	Make various sizes available, encourage students to bring their own 提供多種尺寸選擇,鼓勵學生帶他們自己的
Safety glasses 安全眼鏡	1	Make various sizes available, encourage students to bring their own 提供多種尺寸選擇,鼓勵學生帶他們自己的

8. Appendices 附錄

Appendix A: Activity Preparation Guide 附錄 A: 活動準備指南

IMPORTANT SAFETY NOTICE: 重要安全注意事項

This course includes activities which involve the learner interacting with live electricity and hand tools. It is the sole responsibility of the AST to ensure that:

本課程包含的活動讓學員與現場電力設備及手動工具有所互動, 講師最主要的責任即是確保以下注意事項:

- All Learners are properly supervised at all times while working with live electricity and hand tools
- **所有的學員在都在適當的監督下操作電力設備及手動工具**
- All Learners use appropriate Personal Protective Equipment during course activities
- **所有學員在課程活動期間皆使用適當的個人防護配備**
- All working stations are set up to meet the safe workplace recommendations detailed in guidebook sections 1.01 and 1.06 and all local workplace safety standards
- **所有工作站的設置皆須達到指引手冊章節 1.01、1.06 所載明之安全建議及所有當地工作場域安全標準**
- All Learners are introduced to the safe use of all hand and measurement tools to be used prior to beginning
- **在課程開始前,向所有學員介紹如何安全使用所有手動及測量工具**
- If unsafe behavior is observed at any time:
- **無論何時, 學員一旦違反安全規定,**
- the activity is stopped for the Learner
- **應立即中止該學員的活動,**
- the Learner is provided with feedback
- **並給予學員反饋,**
- the Learner demonstrates an understanding of safe behavior before continuing
  請學員確實理解何謂安全行為後始可繼續參與活動

- All course activities are conducted in accordance with local safety regulations and best practices
- **所有課程活動皆按照當地安全規定及最佳實務進行。**
- All stations which involve live electricity are connected in accordance with local electrical code and workplace safety regulations
- **所有需用電之工作站皆須依照當地電工標準及工作場域安全規定進行連接。**
- All stations which involve live electricity are connected to power using GFCI devices which have been tested for proper function before beginning this exam.
- **所有需用電之工作站連接到電源皆須使用漏電斷路設備，並在測驗前通過功能測試。**
- All outlets and devices are properly grounded and tested by the AST to confirm proper grounding before beginning this course
- **所有插座及設備在程課開始前皆確實接地，並經由講師測試及確認。**
- All hand tools and measurement devices are inspected for safe and proper function before beginning this course
- **所有手動工具及測量設備在課程開始前皆經測試後確認安全性及功能正常。**

If local workplace safety regulations do not allow for a prescribed activity, the AST is not to conduct this activity and is to contact SCA Education to discuss alternative activities

假如當地安全電工標準或工作場域安全規定不允許進行指定的活動，講師將不進行該活動並連絡 SCA 教育團隊討論替代方案

Preventive Maintenance Checklist

預防性維護檢核表

The Activities and Practical Exam in this course include tasks which are commonly performed during a preventive Maintenance visit. To receive full credit on the Practical Exam the Learner must record all observations, calculations, and task completions on the preventive Maintenance Checklist found on the last page of the Practical Exam Learner Worksheet.

In order to ensure the Learners are familiar with this checklist prior to the Practical Exam, the Trainer must provide this checklist during the Activities below. The Trainer may additionally choose to create a more complete sample checklist for the Learner’s future reference, however this checklist should not be used during these Activities or the Practical Exam.

課程中的活動及實務測驗包含了預防性維護服務中一般會執行的任務。為了在實務測驗中得到滿分，學員需在實務測驗學員工作表的最後一頁-預防性維護檢核表上紀錄所有的觀察、計算、及完成任務。

為了確保學員在實務測驗前熟悉預防性維護檢核表，講師必需在下列活動時提供檢核表。

講師可能會額外選擇更完整的檢核表以利學員將來進一步的參考，然而這份檢核表不應在活動或實務測驗中使用。

<table>
<tr><th>Activity Code
活動代碼</th><th>Preparation Guide
準備指南</th></tr>
<tr><td>1.02
Water Treatment System Components
水處理系統零件</td><td>In this activity, learners interact with and learn to identify physical components from sanitary quick-change cartridge and drop-in cartridge style water treatment systems. Prepare common styles and connect supply lines, being sure to include examples of pressure gages and shut off valves of types commonly used regionally. Systems and components should be dry to enable easy assembly and disassembly. Ensure that the students interact with the drop-in cartridge style system that will be used in Activity 1.07.03 and the Practical Exam. This can be done throughout the course or at one specific activity.
Estimated activity time: 15 minutes
在這個活動中，學員將會學習辨認乾淨的快拆濾芯及置入式水處理系統的實體零件並與之互動。準備常見的型式並接上水源，確認樣本中包含區域常見的壓力表及關閉閥。系統及零件應該是乾燥的，以便可以容易的組裝及拆開。確認學員與活動 1.07.3 及實務測驗中的置入式濾芯系統有所互動，這可以透過課程或一個指定的活動來達成。
預估活動時間:15 分鐘</td></tr>
<tr><td>1.06
Measuring Water Composition
測量水的組成</td><td>In this activity, learners practice using several methods to assess the composition of water samples. Trainer must explain, demonstrate, and guide learners through the following tasks:
● Measure Electrical Conductivity and estimated TDS of water using a handheld meter and record on a PM Checklist
● Measure total hardness and alkalinity of water using a drop-count titration test kit and record on a PM Checklist
● Measure pH of water using a handheld meter
● Measure Chlorine level of water using a test strip
● Perform calculation to convert total hardness and alkalinity results from Degrees of hardness or grains per gallon into ppm CaCO3.
● Record total hardness and alkalinity measurements on the SCA Water Chart and assess whether results are within SCA Core Zone recommendation.
● Trainer must create and provide a visual reference for using the drop-count titration test kit which shows color change expected for each test and conversion factor for ppm CaCO3 calculation. This visual aid is to be made available during this activity and the Practical Exam.
Estimated activity time: 45 minutes
在這個活動中，學員練習使用幾種方式以取得水樣本的組成。講師必需透過以下任務解釋、示範及教導學員:
• 使用手持式儀器測量導電度、估測水樣本中的 TDS 並紀錄在預防性維護檢核表上
• 使用滴定測試組合測量總硬度及鹼度並紀錄在預防性維護檢核表上
• 使用手持式儀器測量水樣本的 pH 值
• 使用檢測試紙來測量水樣本的氯含量
• 執行計算以便將總硬度及鹼度由測量結果的德制硬度或克/加崙轉換成 ppm CaCO3.
• 不論測量結果是否在 SCA 水質核心區的建議內，在 SCA 水質圖上紀錄總硬度及鹼度的測量結果</td></tr>
</table>

	• 講師必需提供標示出各個滴定試驗所預計的顏色變化之視覺參照和 ppm CaCO3 的換算係數。此視覺參照供學生於本活動及實務測驗中使用。 **預估活動時間:45 分鐘**
1.07.03 **Maintaining a Drop-in Cartridge System** **維護置入式濾芯系統**	In this activity, learners practice replacing the cartridge in a drop-in style water treatment system. Trainer must explain, demonstrate, and guide learners through the following tasks: ● Turn off water supply to the system ● Release system pressure ● Unscrew the housing from the system head ● Remove the spent cartridge and o-ring (if used) ● Clean and sanitize the housing ● Install a new cartridge into the housing ● Lubricate and install an o-ring into the housing (if used) ● Reassemble housing to the system head ● Restore water supply to the system ● Flush the system as recommended ● Check housing seal for leaks ● Record the installed and change date **Estimated activity time: 15 minutes** 在這個活動中，學員練習更換置入式水處理系統的濾芯。講師必需透過列的任務解釋、示範並教導學員: • 將系統的水供給截斷 • 釋出系統壓力 • 將濾水器外殼自濾頭上鬆開 • 將用過的濾芯及 O 環移除 • 清潔並消毒濾水器外殼 • 將新濾芯裝進濾水器外殼 • 潤滑並將 O 環放回濾水器外殼 • 將濾水器外殼鎖回系統濾頭 • 恢復系統水供給 • 按建議以水沖洗系統 • 確認濾水器外殼鎖緊無洩露 • 紀錄此次濾芯更換日期及下次預計更換日期

	預估活動時間:15 分鐘
1.08 **PM Components** **預防性維護零件**	In this activity, learners interact with and learn to identify physical components from espresso machines, coffee batch brewers, and coffee grinders which are often recommended to be maintained or inspected during a Preventive Maintenance service. Components shall be displayed within a demonstration batch brewer, espresso machine, and espresso grinder, with all equipment disconnected from power supply. Additional samples of components should also be displayed individually outside of the equipment. Ensure that the students interact with the equipment which will be used in Activity 1.13 and the Practical Exam. This can be done throughout the course or at one specific activity. 在這個活動中，學員將會學習辨認預防性維護服務中一般建議維護或檢查的義式咖啡機、滴濾式咖啡機與咖啡磨豆機的實體零件並與之互動。零件在展示時應該在示範的滴濾式咖啡機、義式咖啡機及義式咖啡磨豆機內，所有設備皆不得接上電源。額外的零件樣本也應該個別地在設備外展示。確保學生與活動 1.13 及實務測驗中所使用設備有所互動，這可以透過課程或一個指定的活動來達成 The components to be identified include 應被辨識的零件包含:

Espresso machine brewing group: 義式咖啡機沖煮頭: - Group head 沖煮頭 - Brew valve* 沖煮閥 - Portafilter locking bezel * 沖煮把手固定環 - Group gasket 沖煮頭墊圈 - Dispersion disc* 分水塊 - Group screen 沖煮頭分水網 - Group screw* 分水網螺絲 - Portafilter 沖煮把手 - Portafilter basket	Espresso machine steam valve: 義式咖啡機蒸氣閥 - Valve body 閥門本體 - Valve stem 蒸氣閥頂桿 - Valve seal 閥體墊片 - Valve o-rings 閥門 O 環 - Valve spring 閥門彈簧 - Valve handle 閥體扳桿	Coffee grinder: 咖啡磨豆機 - Bean hopper 儲豆槽 - Grinding chamber 研磨腔室 - Grinding burrs 刀盤 - Burr carrier 刀座 - Burr carrier shear plate* 刀座插銷 - Outlet chute 出粉通道 - Declumping device* 防結塊裝置 Note: show components representing both espresso and filter coffee grinders 注意: 展示零件應能夠代表義式及一般單

<table>
<tr>
<td></td>
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<tr>
<td>沖煮把手濾杯
- Portafilter basket retaining spring
濾杯固定彈簧</td>
<td></td>
<td>品磨豆機</td>
</tr>
<tr>
<td>Espresso machine steam boiler:
義式咖啡機蒸氣鍋爐:
- Pressure safety valve
壓力安全閥
- Anti-vacuum valve
真空排除閥
- Water level probe
水位探針
- Pressure switch*
壓力開關</td>
<td>Ball-end style espresso machine steam wand joint assembly:
義式咖啡機球型接頭蒸氣棒
- Steam joint nut
蒸汽管固定接頭螺帽
- Steam wand ball-end
蒸氣棒球型接頭
- Spring
彈簧
- Spring bushing
彈簧襯套
- O-ring
O 環
- Gasket
墊片</td>
<td>Batch brewer:
滴濾式咖啡機
- Water level probe
水位探針
- Spray head
噴灑頭</td>
</tr>
</table>
*Note that equipment designs vary and not all equipment will exhibit all components indicated. Components not presented on demonstration equipment may be represented by pictures.
* 注意設備的設計是多樣的，並非所有的設備都會有標示的所有零件。未在示範設備中展示的零件需以照片形式來說明。

Estimated activity time: 30 minutes
預估活動時間:30 分鐘
</td>
</tr>
<tr>
<td>1.11.02
PM Checklist
預防性維護檢核表</td>
<td></td>
</tr>
<tr>
<td>1.11.03
Backflushing an Espresso Machine Brew Group
義式咖啡機沖煮頭的逆洗</td>
<td>In this activity, the Trainer demonstrates steps recommended for backflushing an espresso machine brew group using detergent.
在這個活動中,講師示範使用清潔劑逆洗義式咖啡機沖煮爐頭的建議步驟
Estimated activity time: 5 minutes
預估活動時間:5 分鐘</td>
</tr>
</table>

1.11.04 - 1.11.05 **Espresso Machine PM** **義式咖啡機的預防性維護**	In this activity, learners practice recommended maintenance and inspection steps for espresso machine PM service. Ensure that the Learners interact with the same or identical espresso machine that will be used in the Practical Exam. Trainer must explain, demonstrate, and guide learners through the following tasks: ● Replace group screen and gasket ● Rebuild a ball-end style steam wand joint ● Inspect an espresso machine for proper function ● Record service activities on a PM checklist 在這個活動中,學員練習義式咖啡機預防性維護服務中建議的維護及檢查。確認學員與實務測驗中所使用同型或同一台的義式咖啡機有所互動。講師必需透過以下任務解釋、示範及教導學員: ● 更換爐頭分水網及墊圈 ● 更換球型蒸氣棒接頭 ● 檢查義式咖啡機的正常功能運作 ● 在預防性維護檢核表上紀錄下服務活動 **Estimated activity time: 45 minutes** **預估活動時間:45 分鐘**
1.11.06 **Batch Brewer PM** **滴濾式咖啡機的預防性維護**	In this activity, learners practice inspecting and cleaning a batch brewer fill level probe. Ensure that the Learners interact with the same or identical batch brewer that will be used in the Practical Exam. Trainer must explain, demonstrate, and guide learners through the following tasks: ● Confirm the brewer is disconnected from electrical power and remove necessary panels ● Remove, inspect, and clean level probe ● Reinstall level probe and reassemble panels 在這個活動中，學員練習檢查並清潔滴濾式咖啡機的水位探針。確認學員與實務測驗中所使用同型或同一台的義式咖啡機有所互動。講師必需透過以下任務解釋、示範及教導學員: ● 確認滴濾式咖啡機並無接上電源並移除必要的隔板 ● 拆除、檢查並清潔水位探針 ● 重新裝上水位探針並將隔板重新組裝恢復 **Estimated activity time: 15 minutes** **預估活動時間:15 分鐘**
1.11.07 **Espresso Grinder Preventive Maintenance**	In this activity, learners practice replacing espresso grinder burrs. Ensure that the Learners interact with the same or identical espresso grinder that will be used in the Practical Exam. Trainer must explain, demonstrate, and guide learners through the following tasks: 在這個活動，學員練習更換義式咖啡磨豆機的刀盤。確認學員與實務測驗中所使用同型或同一台的義式咖啡磨豆機有所互動。講師必需透過以下任

義式咖啡磨豆機的預防性維護	務解釋、示範及教導學員: • Disconnect grinder from electrical power • Open grinding chamber and removes worn burrs • Thoroughly clean grinding chamber and burr carriers • Install new grinding burrs using an appropriate screw tightening procedure • Reassemble grinder • Calibrates grinder and adjusts to an appropriate setting • 斷開磨豆機的電源 • 打開研磨腔室並移除舊的磨損刀盤 • 徹底清潔研磨腔室及刀座 • 換上新的刀盤並以正確地程序栓緊螺絲 • 重新組裝恢復磨豆機 • 校正磨豆機並調整至適當的設定 **Estimated activity time: 30 minutes** **預估活動時間:30 分鐘**

Appendix B: Glossary of Terms

附錄 B: 專業術語

Course Term **課程專業名詞**	**Definition or Description** **定義或描述**	**Alternative Terms** **替代術語**	**Reference** **相關資源**
Alkalinity 鹼度	The ability of the water to neutralize acids. Also describes the concentration of carbonate and bicarbonate compounds in the water. 水能夠中和酸的能力。也描述水中碳酸根及碳酸氫根化合物的濃度		SCA Water Quality Handbook, pg. 77
Anion 陰離子	Negatively charged ion. 帶負電的離子		SCA Water Quality Handbook, pg. 77
Anti-vacuum valve 真空排除閥	A mechanical component that allows air to enter a steam boiler or circuit to break a vacuum that may occur as the chamber containing the steam cools. 當鍋爐中內含蒸氣冷掉後，讓空氣進入蒸氣鍋爐或破壞真空的機械性零件	Vacuum breaker 真空洩壓閥	
Backflush 逆洗	A procedure for rinsing or cleaning espresso machine brewing groups by causing water or detergent solution to flow backwards from a blank portafilter basket, through the shower screen and brewing valve, and out through the brew valve exhaust port. 藉由讓水或清潔溶劑從無孔濾杯回流經分水網、沖煮閥而由沖煮閥排水口而出的方式，達到沖洗或清潔義式咖啡機沖煮頭的程序		
Ball-end joint 球型接頭	A pivoting joint, often found on the end of a steam or hot water wand, consisting of a rotating spherical surface which is pressed against a concave metal surface or polymer gasket by a spring and spherical bushing to create a leak-proof seal. 一個旋轉接頭，通常在蒸氣棒或熱水棒的末端，由一個旋轉球型面透過一個彈簧及球型襯套壓緊在金屬凹面或聚合物墊片上，創造出防漏的密封狀態		
Bar 大氣壓力	A unit of measurement for pressure as related to the atmosphere. One bar is Equal to (1) atmosphere or 14.7PSI(G)		

	一個與大氣有關測量壓力的單位。一個 bar 等於一大氣壓力或 14.7PSI(G)		
Bean hopper 咖啡豆槽	A coffee grinder component that holds coffee beans to be ground 將待磨咖啡豆托存住的一個咖啡磨豆機零件		
Boiler level sight glass 鍋爐水位視窗	A glass component that allows the user to see a representation of the current water level in a steam boiler. 讓使用者可以看到蒸氣鍋爐中目前水位表示的一個玻璃零件		
Boiler safety valve 鍋爐安全閥門	A safety component mounted to a steam boiler that relieves excess pressure. 一個安裝在蒸氣鍋爐的安全零件，可以釋放出多餘的壓力		
Brew pressure gauge 沖煮壓力表	An instrument that measures and displays the pressure of the water being used for espresso extraction. 測量並顯示義式咖啡萃取時水壓的儀表		
Brew valve 沖煮閥	An electro-mechanical valve that opens to allow the flow of water to the coffee bed when a brew cycle is activated. 一個電子機械閥，當沖煮循環啟動時，打開時讓水流經咖啡粉餅	Three-way valve 三通閥	
Burr carrier 刀座	A coffee grinder component used to mount a grinding burr. Grinders typically contain one stationary burr carrier and one moving burr carrier, often mounted to the output shaft of a motor. 一個安裝於磨豆機刀盤的咖啡磨豆機零件。磨豆機通常包含一個固定的刀座和一個移動的刀座，通常安裝在馬達的輸出軸上		
Burr carrier shear plate 刀座插銷	A coffee grinder component which transmits torque from the grinder motor output shaft to the moving burr carrier under normal use but is designed to fail if excessive loads are encountered in order to minimize damage to the grinder. 一個咖啡磨豆機零件，在正常使用下，由磨豆機馬達輸出軸傳遞扭力至移動的刀座，但為了減少對磨豆機的損害，在多餘的負載發生時會失去功效		
Carbonate	Carbonate hardness effectively corresponds to the maximum amount of scale that can	Temporary	SCA Water Quality Handbook, pg. 77

Hardness 碳酸鹽硬度	form and is therefore equal to whichever of the two values of total hardness and alkalinity is lower. Synonymous with Temporary Hardness. 碳酸鹽硬度實際上相當於結垢的最大量，因此等同於總硬度及鹼度二者中比較低的，為暫時硬度的同義詞	Hardness 暫時性硬度	
Cation 陽離子	Positively charged ion. 帶正電的離子		
Chloride 氯離子	The chloride ion is the anion Cl . It is formed when the element chlorine (a halogen) gains an electron or when a compound such as hydrogen chloride is dissolved in water. 氯離子即是陰離子 CL，它的形成來自於氯元素得到了一個電子或當像氯化氫的化合物溶解於水中時		Chloride - Wikipedia
Chlorine 氯	Tap water is chlorinated with hypochlorite (Cl2) to kill bacteria and microbes. Not to be confused with chlorides such as sodium chloride (NaCl), calcium chloride (CaCl2), or magnesium chloride (MgCl2). 自來水使用次氯酸鹽的氯化來消滅細菌及微生物，不要與氯化鈉(NaCl)、氯化鈣(CaCl2)或氯化鎂(MgCl2)中的氯離子混淆		SCA Water Quality Handbook, pg. 78
Compound 化合物	An electrically neutral group of two or more atoms of the same or different elements, held together by chemical bonds (e.g. Oxygen (O2); Water - two hydrogen atoms bonded to an oxygen atom (H2O); or Salt (NaCl). 一個有著二個或多個一樣或不同元素的不帶電的原子團，透過化學鍵綁在一起(例:氧(O2)、水-二個氫原子鍵合一個氧原子(H2O)或鹽(NaCl).)		SCA Water Quality Handbook, pg. 78
Decarbonizer 脫碳劑	Decarbonization is also an ion-exchange method, though in contrast to the former, the calcium and magnesium ions are exchanged for protons (H+). The released protons in turn neutralize hydrogen carbonate by formation of carbonic acid. Decarbonization leads to a decrease in total hardness and alkalinity by equal amounts. 脫碳劑也是一種離子交換的方法，與之前相比較，鈣離子及鎂離子與氫離子(H+)交換。釋放出的質子反過來中和碳酸氫鹽形成碳酸。脫碳使總硬度及鹼度以相同	Decarbonizing Softener, alkalinity Softener 脫碳軟化, 去鹼度軟化	SCA Water Quality Handbook, pg. 78

	量值的減少。		
Declumping device 防結塊裝置	A coffee grinder component which breaks up clumps of coffee grounds in the grinder output chute. 咖啡磨豆機中的裝置，在磨豆機的出粉通道中破除咖啡粉的結塊		
Dispersion Plate 分水塊	The part of the group head assembly that distributes water in order to shower over the dispersion screen. This is also referred to as a shower plate. 咖啡沖煮頭中一部分的組合體，將水以花灑狀分配到分水網上，也被稱為花灑盤	Dispersion disc 分水片	
Drain tube 引流管	Tubing dedicated to connecting the drain manifold to the drain system of a facility or a dedicated receptacle. 專用來連接排水歧管至某一個裝置的排水系統或專門容器的管道		
Electrical Conductivity 導電度	Electrical conductivity (EC) measures the ability of a substance to transmit or conduct electricity. Since pure water is a pure conductor, EC in drinking water originates mainly from dissolved ions. Electrical conductivity has often been used to estimate the total dissolved solids content (TDS) of water samples but conversion factors can be up to +/-50%. 導電度測量某一物質傳遞或導電的能力。因為純水是幾乎不導電的，飲用水中的導電度主要來自於溶解的離子。導電度也常用來估測水樣本的總溶解固體含量(TDS)，但轉換因子至多可到達+/-50%。		SCA Water Quality Handbook, pg. 78
Expansion valve 膨脹閥	A component that relieves pressure from brew boilers at a set point, which is often 12 bars. May or may not be field-adjustable. 這個零件在來自沖煮鍋爐的壓力達到一個設定點，通常是 12 個 Bar 時會釋放壓力。可能也可能不能調整範圍		
Flowmeter 流量計	A measurement device that tracks the volume of liquid passing through it and sends information to a control board.		

	追踪通過它的液體體積並傳送訊息給控制板的量測設施		
Grains per Imperial Gallon 格令/英制加崙	A unit of water hardness defined as 1 grain (64.8 milligrams) of calcium carbonate dissolved in 1 US gallon of water (3.78 L). It translates into 1 part in about 58,000 parts of water or 17.1 parts per million (ppm). Also called Degrees Clark. 一個水的硬度單位，定義為 1 格令(64.8 毫克)的碳酸鈣溶解在 1 美式加崙的水(3.78L)。它可以轉化為 1 等份對大約 58000 等份的水或 17.1 等份每百萬(ppm)。也稱為克拉克硬度	Degrees Clark 克拉克硬度	SCA Water Quality Handbook, pg. 78
Grinding burr 研磨刀盤	A coffee grinder component, typically made from hardened steel or ceramic material, and featuring a large number of teeth which are used to fracture coffee beans into progressively smaller particles. 咖啡磨豆機的零件，通常由硬化鋼或陶瓷材料所製成，特質上有著大量用來使咖啡豆進一步破碎成更小的顆粒的牙	Grinding blade	
Grinding chamber 研磨腔室	An enclosure within a coffee grinder which contains the grinding burrs, directs the flow of whole coffee beans into the burrs, and directs the flow of ground coffee particles into an outlet chute. 咖啡磨豆機內一個封閉區域，內含研磨刀盤、將整顆咖啡豆送至刀盤，並將研磨後的咖啡顆粒送進出粉通道。		
Group/group Head 沖煮頭	An assembly of components that receives the portafilter and delivers brewing water to the coffee bed at a preset temperature and flow rate, AKA brew unit/pouring unit. 接上把手並將沖煮水以預定的溫度及流速送到咖啡粉餅上的零件組合體，也稱為沖煮元件	Brewing group 沖煮頭	
Hardness 硬度	Now referred to as 'total hardness', this is the concentration of calcium (Ca) and magnesium (Mg) in the water. More calcium and magnesium = harder water. Less calcium and magnesium = softer water. 現在稱為總硬度，這是水中鈣與鎂的濃度。越多的鈣與鎂=較硬的水。越少的鈣與鎂=較軟的水		SCA Water Quality Handbook, pg. 79

Ion 離子	Atoms or molecules that have a net positive or negative charge. 有著淨正電荷或淨負電荷的原子或分子		SCA Water Quality Handbook, pg. 79
Langelier Scaling Index 朗格利爾飽和指數	An approximate indicator of the degree of saturation of calcium carbonate in water. It is calculated using the pH, alkalinity, calcium concentration, total dissolved solids, and water temperature of a water sample collected at the tap. 水中碳酸鈣飽合度的一個大約指標。它的計算使用自來水水樣本的 pH、鹼度及鈣濃度，總溶解固體含量和水溫		SCA Water Quality Handbook, pg. 79
Larson-Skold-Index 拉森指數	A measure for the corrosivity of water towards steel. It is calculated by dividing the sum of chloride and sulfate by alkalinity. 水對鋼材侵蝕性的量測，計算使用氯及硫酸鹽的加總除以鹼度		SCA Water Quality Handbook, pg. 79
Limescale 結垢	The hard white chalky deposit found in kettles and boilers. It originates from limestone that is dissolved from the ground and then re-forms later in equipment when the water chemistry changes. It is mostly calcium carbonate ($CaCO_3$), but in many places magnesium carbonate ($MgCO_3$) may account for up to 1/3 of the total. 在水壺或鍋爐中所找到白色硬粉狀的沈澱物，來自於地層中石灰岩的溶解物，然後因水質的化學改變，在設備中再次成形。它多數來自於碳酸鈣($CaCO_3$)，但在很多地方，碳酸鎂可能至多會佔其中的 1/3		Equilibrium in Water and Coffee, pg. 32
Locking bezel 固定框架	Ring-shaped portion of the brewing group with tapered slots which accept the tabs on the sides of a portafilter and are designed to press the basket rim against the group gasket when locked into place to create a seal. 環形部分的沖煮頭，有著變窄的狹槽能夠讓沖煮把手二側的襟片卡上，並被設計為將濾杯的邊緣壓在沖煮墊片上而鎖進位置上創造出密封狀態		
Maintenance interval 保養週期	A recommendation for the amount of time a component or system should be used between preventive maintenance service activities. 建議在預防性維護服務之間一個零件或系統的使用壽命		
mg/L	Milligrams per liter. It is the equivalent to ppm when testing water at <45°C.		SCA Water Quality Handbook, pg. 79

毫克/升	毫克每升。當測試溫度<45°C，它等同於 ppm		
Molecule 分子	An electrically neutral group of two or more atoms held together by chemical bonds. For example, Oxygen (O2) or Water (H2O). 一個有著二個或多個透過化學鍵綁在一起的不帶電原子團 例如: 氧(O2)或水(H2O).		SCA Water Quality Handbook, pg. 79
Non-Carbonate Hardness 非碳酸鹽硬度	The concentration of calcium and magnesium occurring as compounds other than carbonates in the water. 鈣鎂在水中以非碳酸鹽化合物存在的濃度		SCA Water Quality Handbook, pg. 79
Not Hard Carbonates 非硬度碳酸根	The concentration of minerals, other than calcium and magnesium, which occur as carbonate compounds. 非鈣鎂的礦物質以碳酸鹽化合物存在的濃度		SCA Water Quality Handbook, pg. 79
Outlet chute 出粉通道	A feature of a coffee grinder which directs the flow of coffee grounds from the grinding chamber into the desired location, typically a portafilter basket, ground coffee funnel, brew basket, coffee bag, or dosing cup. 咖啡磨豆機中引導咖啡粉由咖啡腔室進入預期的位置，通常是沖煮把手的濾杯、咖啡粉料斗、沖煮濾杯、豆袋或接粉杯中		
Permanent Hardness 永久硬度	The calcium and magnesium that cannot precipitate from water when boiled. 當水滾時無法由水中沈澱而出的鈣與鎂		SCA Water Quality Handbook, pg. 79
pH	This describes whether the water is more acid, more alkaline, or neutral. 這描述了水是否是較酸、較鹼或中性		SCA Water Quality Handbook, pg. 79
Polyphosphate 多磷酸鹽	Food grade polyphosphates, in basic phosphorus bonds, react with the lime minerals in the water and thus disturb the scale deposits. 食品等級的多磷酸鹽，以基本的磷鍵結構，與水中的水垢礦物質產生反應，因此擾亂了水垢的沈澱	Scale Inhibitor 阻垢劑	Equilibrium in Water and Coffee, pg. 71

Portafilter 沖煮把手	A removable, handheld component of a lever, semi-automatic, or volumetric espresso machine that performs the function of retaining the filter basket and securing it into the group head during the extraction process. It includes a pouring spout to direct the flow of coffee into a container. 一個可移除、手持式的零件，在拉把式義式咖啡機、半自動義式咖啡機或定量式義式咖啡機萃取的過程中，容納濾杯並將其緊扣至沖煮頭中。它包含了將咖啡液導流至容器中的導流嘴。		
Portafilter Basket 沖煮把手濾杯	A component of a traditional espresso machines that serves as the container for the ground coffee during extraction. The bottom of the basket features finely stamped or machined holes that allow the extracted coffee to exit while retaining the grounds. 傳統義式咖啡機中在萃取時作為咖啡粉容器的一個零件。濾杯的底部有著許多細小的機器沖孔，讓萃取後的咖啡可以流出卻攔住咖啡粉		
Ppm	Parts per million. It is equivalent to mg/L when testing water at <45°C. 百萬分之一，當測試溫度<45°C，它等同於克/毫升		SCA Water Quality Handbook, pg. 79
Pressure gauge 壓力表	An instrument that measures and displays the current pressure within the steam or coffee boiler on an espresso machine. 一種設備，可以測量並顯示義式咖啡機內的蒸氣或咖啡鍋爐的最近溫度		
Pressure switch 壓力開關	An electrical switch that is activated by a pressure sensitive mechanism to control the power supply to an espresso machine steam boiler heating element. 一種電子開關，由壓力敏感機制所觸發，進而控制義式咖啡機蒸氣鍋爐加熱元件的電力供給	Pressure stat 自動調壓器	
Preventive Maintenance 預防性維護	Regular and routine service performed to maintain proper system function and reduce unexpected equipment failure. 定期常規的服務執行以維持適當的系統功能並減少無法預期的設備故障		
PSI	A unit of measurement for pressure, representing Pounds per Square Inch		

	一種測量壓力的單位，代表每一平方英寸的磅數		
Pump pressure 泵浦壓力	The pressure of a liquid when water is moving through the system. 當水正在穿過系統時的液體壓力		
Reactive Maintenance 反應性維護	Unplanned repair of equipment that has failed in order to restore proper system function. 設備故障後,為了讓系統回到正常功能運作而進行未在計畫內的修復	Emergency service 緊急維修	
Reverse Osmosis 逆滲透	A pressure-driven membrane process used for purification of water. 一種壓力趨動穿過濾膜的過程，用以進行水的純化		Equilibrium in Water and Coffee, pg. 92
Shower screen 分水網	A fine, wire-mesh screen that evenly distributes brew water as it flows into a portafilter. This is also referred to as a dispersion screen. 一個細密的網狀濾網，讓沖煮水得以平均分布地流進把手，英文也稱之為 dispersion screen。	Group screen 沖煮頭分水網	
Shower screw 分水網螺絲	Screw, usually made of stainless steel, which secures the shower screen to the group 將分水網栓進沖煮頭的螺絲，通常為不鏽鋼製	Group screw 沖煮頭螺絲	
Shutoff valve 關閉閥	An inline component that is used to control the flow of water or steam manually. 一個串聯零件，用來手動控制水或壓力的流動		
Sodium (Na) 鈉	Alkali metal that can occur as a chloride compound (e.g. Salt (NaCl)), or as a carbonate compound in water. 鹼金屬，可以以氯化物(例: 鹽(氯化鈉))或以一種碳酸鹽化合物存在於水中		SCA Water Quality Handbook, pg. 80
Softener 軟化器	A softener is an ion-exchange method where calcium and magnesium ions are exchanged for either sodium or potassium ions. This treatment reduces total hardness without affecting alkalinity. 軟化器是一種離子交換的方式，用鈉離子或鉀離子來交換鈣離 子和鎂離子，這樣的處理在不影響鹼度的狀態下減少了總硬度		SCA Water Quality Handbook, pg. 80

Spray head 噴灑頭	Batch brewer component that promotes the uniform distribution of water onto the bed of coffee grounds 滴濾式咖啡機中的零件，促使水均勻的分布進咖啡粉餅中		
Steam boiler 蒸氣鍋爐	A vessel where water is converted from liquid to steam and stored under pressure. 水在這個容器中由液體轉換成蒸氣並在壓力儲存其中		
Steam valve 蒸氣閥	A component on espresso machines that controls the flow of steam from the boiler to the steam wand, either by a control lever, knob, or electro-mechanical solenoid and switch. 義式咖啡機中的零件，控制蒸氣的流動由鍋爐流向蒸氣棒，或由控制桿、旋鈕或機電螺線管及開關		
Steam wand 蒸氣棒	A teflon, stainless or plated brass component used for steaming milk, with a removable tip that releases a controlled blast of steam through one or more small holes. 用來蒸牛奶的一種鐵氟龍、不鏽鋼或鍍黃銅的零件，有著可移除的尖端，可透過一個或多個小洞釋放出一股可控制的蒸氣		
Supply line 供水線	A tube or hose connecting the building's water supply to the machine's inlet/fill valve. 連接建築物供水至機器進水閥的軟管或金屬管		
Temporary Hardness 暫時硬度	The calcium and/or magnesium can precipitate from the water when boiled, and form scale deposits. 當水煮滾後,會從水中沈澱析出的的鈣和鎂，並形成結垢沈積物		SCA Water Quality Handbook, pg. 80
Total Dissolved Solids (TDS) 總溶解固體含量	Total dissolved solids simply means the sum of all solid substances present in the water. For water, this parameter is often estimated by electrical conductivity by assuming a fixed composition. A conversion factor of 0.7 in mg/L of TDS per 1 µS/cm of electrical conductivity is used by the heritage SCAA Water Standard. In laboratory tests, it is measured by evaporation and weighing of the solid residue – also including uncharged dissolved solids such as silicates that do not contribute to electrical conductivity.		SCA Water Quality Handbook, pg. 81

	總溶解固體僅表示水中所有存在的固體物質總和。對水來說，在假設組成固定的狀態下，這個參數通常會以導電度來作估測。換算因子以每 1 µS/cm 的導電度，則有 0.7 毫克/升的總溶解固體，繼承自 SCAA 水質標準的遺產 在實驗室的測試中，它的量測會透過蒸發並對固體殘留物進行秤重，這也包含了不導電的溶解固體，像是矽酸鹽，它們並無法貢獻導電度		
Total Hardness 總硬度	Defined as the sum of calcium and magnesium in equivalent concentrations (or molar concentrations). In rare cases, other ions can contribute to total hardness, for example strontium. 定義為在當量濃度(莫耳濃度)下，鈣與鎂的總和，在極少的案例中，其他離子亦對總硬度有所貢獻，例如像鍶		SCA Water Quality Handbook, pg. 81
Water pump 水泵	A positive-displacement mechanism that traps the liquid entering from an inlet pipe and forces it out through a discharge pipe. It is operated by an attached motor. 一個正排氣型的機制，把液體從進水管中收集起來，而後迫使其由排放管中傾瀉而出。藉由一台附加的馬達而達成運作		
Water-level fill probe 滿水位探針	An electrical conductivity sensor that signals for a boiler to be filled to a desired level. 電子導電度感應器會在鍋爐將滿到預期水位時發出信號		